

A Multi-Tissue Transcriptomic Atlas of Porcine Development Identifies Conserved Molecular Programs with Humans

Tianyuan Liu^{1,2,3}, Rujing Lei¹, Ilyas M. Khan⁴, Peter Theobald^{1,*}

¹*Cardiff School of Engineering, Cardiff University, The Parade, Cardiff, CF24 3AA, UK*

²*Institute for Integrative Systems Biology (I2SysBio), Spanish National Research Council (CSIC), Paterna 46980, Spain*

³*Department of Computer Science, University of Valencia, Spain*

⁴*Faculty of Medicine, Health & Life Science, Swansea University Medical School, UK*

*Corresponding author: TheobaldPS@Cardiff.ac.uk

Abstract

Background: Experimental reproducibility and cross-species translation using the domestic pig (*Sus scrofa*) are limited by the lack of a standardised molecular framework to define biological maturation. While the pig is an excellent preclinical model for human conditions, its developmental tempo differs substantially from humans, and no tissue-resolved transcriptomic staging system exists. Synchronising the maturation rates of *Sus scrofa* and humans is essential to move preclinical research from a descriptive exercise to a predictive science.

Results: We developed a multi-tissue transcriptomic atlas mapping porcine development across five major tissues (muscle, brain, liver, blood, and lung) using 1,924 RNA-seq profiles from the PigGTEx resource. Our ordinal machine learning framework achieves tissue-specific developmental staging with a mean balanced accuracy of 0.90 by nested cross-validation, and reveals highly tissue-specific molecular programs underlying maturation (cross-tissue feature overlap: 0.2% Jaccard similarity). Cross-species comparison with human skeletal muscle development ($n = 11$) shows 97% directional concordance (64/66 genes) and a significant correlation ($r = 0.69$, 95% CI: 0.54–0.80) among 66 conserved developmental genes organised into three functional modules: neuromuscular maturation, structural refinement, and metabolic transition. An interactive web application for developmental stage prediction is publicly available at <https://pigdevstage.streamlit.app>.

Conclusions: This transcriptomic atlas provides the first standardised molecular framework for porcine developmental staging, enabling precise synchronisation of experimental cohorts. The conserved pig–human developmental programs observed in muscle support the conclusion that porcine

molecular maturation shares core biological mechanisms with humans, providing a framework for more accurate cross-species developmental comparisons in nutrition, physiology, and disease modelling.

Keywords: Porcine development, Transcriptomic atlas, Developmental staging, Machine learning, Cross-species conservation, Gene expression, Ordinal classification, PigGTEx

1 Background

The domestic pig (*Sus scrofa*) is central to modern animal science, serving as both a major agricultural species and an increasingly important model organism[1–3]. However, a critical gap undermines reproducibility across pig research: the lack of a standardised, molecular definition of developmental stage. Current staging relies on morphological criteria or chronological age, which are poor proxies for biological maturity[4]. Littermates frequently develop asynchronously, and distinct organs mature at different rates, effectively decoupling whole-body age from tissue-specific functional competence. This imprecision confounds experimental reproducibility in nutrition, growth physiology, and disease studies, where developmental stage is a critical variable[5].

Molecular approaches offer a solution to this challenge. A ‘universal’ epigenetic clock trained on 185 mammalian species[6], together with porcine-specific clocks[7], now enables broad translation of biological age between humans and other mammals. Although tissue-specific methylation clocks can be constructed[8–10], they primarily capture the gradual, cumulative drift of CpG methylation with age rather than the abrupt transcriptional switches that mark developmental phase transitions[11]. Epigenetic age can therefore align two species globally while organ-level maturation remains asynchronous—a pig is locomotor-competent at birth, yet its immune and metabolic systems lag behind the corresponding human timeline. Transcriptomic profiling complements methylation-based clocks by providing a direct, functional readout of cellular state, capturing phase-specific gene programs that define tissue maturation[12, 13]. Moreover, whereas the biological interpretation of individual clock CpGs remains largely unclear—most epigenetic clocks reflect composites of cell division, DNA damage, metabolic activity, and cell-type composition changes[10]—transcriptomic features are directly interpretable as active gene programs. Transcriptomic profiling thus enables the construction of *developmental clocks* that are mechanistically transparent and tuned to the functional programs defining organ-specific maturation phases. Yet a comprehensive, tissue-resolved developmental atlas integrating such transcriptomic data has been missing for the pig[14, 15], limiting the standardisation of experimental protocols across research groups.

Here, we present the first multi-tissue transcriptomic atlas of porcine development, integrating data across five major tissues (brain, liver, muscle, lung, and blood) with sufficient sample sizes for machine learning analysis. Unlike previous descriptive studies, we build a quantitative, predictive framework

that infers developmental stage with high accuracy directly from gene expression profiles. To assess the biological relevance of our atlas, we performed a cross-species comparison with human skeletal muscle ($n = 11$), finding a significant positive correlation of developmental fold-changes (Pearson $r = 0.69$, $n = 66$ genes, 97% directional concordance) and identifying shared regulatory modules involved in postnatal maturation in both species. We note that cross-species validation is currently limited to muscle tissue due to availability of matched human developmental data. This work establishes a rigorous molecular standard for staging in pig research, accompanied by a publicly accessible web application (<https://pigdevstage.streamlit.app>), enabling precise synchronisation of experimental cohorts and improving reproducibility across studies in swine nutrition, production, and biomedical applications.

2 Results

2.1 A comprehensive transcriptomic atlas of porcine development

To define a molecular standard for porcine development, we constructed a large-scale reference atlas using RNA-seq profiles from the PigGTEx resource[16–18], a pig analogue of the human GTEx project[19] (Fig. 1a). The dataset comprises 1,924 samples spanning five major tissues (brain, liver, muscle, lung, and blood) and covering the postnatal lifespan from the day of birth to beyond 365 days of age; prenatal (embryonic and fetal) development is outside the scope of the present atlas, since the PigGTEx resource does not include systematic, multi-tissue prenatal sampling and addressing this developmental window would require dedicated prenatal sample collection (see Limitations). These five tissues were selected for machine learning analysis based on postnatal sample availability and developmental stage representation.

We addressed the limitations of chronological age by defining five life stages based on physiological milestones: infant (0–20 d), early childhood (21–59 d), pre-pubertal (60–149 d), post-pubertal (150–365 d), and adult (>365 d) (Fig. 1c). These stages correspond to distinct physiological transitions grounded in porcine developmental biology[1]: the infant period captures the pre-weaning neonatal phase, during which the gastrointestinal transcriptome undergoes intensive remodelling[20]; early childhood spans the weaning transition (typically 21–28 d) and its associated gut adaptation[5]; the pre-pubertal stage encompasses rapid juvenile growth following weaning completion (~ 60 d)[21]; the post-pubertal boundary at 150 d marks the earliest onset of puberty in gilts (reported range: 150–220 d depending on genetics and management[1, 22]), chosen to capture the molecular preparation for sexual maturation that precedes behavioural oestrus[22]; and adulthood (>365 d) represents full reproductive and physiological maturity. This provided a ground truth framework for training, allowing us to decouple “biological time” from simple chronological time. The resulting dataset serves as the most comprehensive transcriptomic resource for porcine development to date.

2.2 Accurate inference of physiological developmental stage

We implemented a machine learning framework to infer the biological developmental stage directly from tissue-specific transcriptomes (Fig. 1b). Models were evaluated using nested cross-validation with automatic hyperparameter tuning to ensure unbiased estimation (see Methods). Pooled held-out performance demonstrated robust generalisation across tissues, with a mean balanced accuracy of 89.5% across all tissues (Fig. 2a; see Supplementary Table S1 for detailed metrics).

Performance varied according to tissue complexity and sample availability. Blood achieved the highest balanced accuracy (95.9%), followed by liver (92.1%), muscle (89.4%), and lung (89.2%), whereas the brain, with its extreme cellular heterogeneity and limited sample size ($n = 43$), showed lower predictability (81.1%). Notably, tissues with severe class imbalance required reduced classification schemes (4-class for muscle and liver, 2-class for brain, blood, and lung; Fig. 1d) to ensure robust model training (Supplementary Table S1; Supplementary Fig. S3a). Consequently, the atlas provides high-resolution multi-stage developmental classification for muscle and liver, but only binary maturity classification (pre-pubertal vs. post-pubertal) for brain, blood, and lung; finer staging for these tissues will become feasible as more balanced developmental data accumulates.

Crucially, the system achieved near-perfect ordinal consistency in well-sampled tissues like muscle ($\rho = 0.965$) and liver ($\rho = 0.949$) (Fig. 2c). Inspection of the per-tissue confusion matrices (Fig. 2b) confirms that residual misclassifications are concentrated almost exclusively between *adjacent* developmental stages—for example, infant samples occasionally classified as early childhood, or pre-pubertal samples classified as post-pubertal—rather than being distributed randomly across the ordinal scale. Together with the per-stage precision/recall/ F_1 heatmap (Supplementary Fig. S3a) and per-class values reported in Supplementary Table S2, these patterns indicate that the most difficult decision boundaries lie at the genuinely continuous transitions of postnatal biology (notably the early-childhood-to-pre-pubertal transition, where post-weaning physiology overlaps with juvenile growth), rather than at boundaries of equal biological difficulty.

These tissue-specific performance differences prompted us to ask: what biological programs underlie the model’s predictions? To answer this, we examined the functional annotation of the genes that drive accurate stage inference.

2.3 The functional landscape of development

Beyond prediction, our model uncovers the biological logic driving maturation. We performed functional enrichment analysis on the top informative genes for each stage, mapping the global landscape of developmental regulation (Fig. 3).

The resulting functional network reveals that each tissue follows a distinct regulatory framework. In the brain, development is driven by synaptic refinement and metabolic support[23, 24], highlighted by

enrichment of *sarcosine metabolism* (linked to NMDA receptor coagonist availability[25]) and *beta-tubulin binding* (critical for axonal transport). In contrast, the liver prioritises metabolic autonomy and homeostasis, evidenced by the upregulation of *autophagosome-membrane adaptors*. Muscle development centres on contractile machinery assembly and neuromuscular signalling (e.g., *acetylcholinesterase activity*[26]), while the lung focuses on immune system establishment[27] (*IL-6 signalling regulation*) and RNA metabolic remodelling (*m6A reader activity*). This systems-level view moves beyond simple marker genes to capture the coordinated, tissue-specific logic of physiological maturation.

Notably, feature importance rankings showed moderate instability across random seeds (mean Jaccard similarity: 0.08–0.31 across tissues; Supplementary Fig. S2b), with 6–42% of top 50 features appearing in $\geq 80\%$ of runs. This is consistent with the known behaviour of tree-ensemble methods when input features are highly correlated: many distinct feature subsets from co-regulated gene modules can yield equivalent predictions. Importantly, we distinguish between *feature instability* and *model instability*: while the specific genes selected vary across seeds, prediction accuracy is highly stable (e.g., muscle: BA mean = 0.88, std = 0.03 across seeds), confirming that the model captures robust developmental patterns regardless of which specific genes from correlated modules are selected. The selected features showed strong correlations with developmental stage (mean absolute $\rho = 0.2$ – 0.6 ; Supplementary Fig. S2c). Critically, cross-tissue feature overlap was extremely low (0.2% Jaccard similarity). This profound tissue-specificity confirms that each tissue uses unique molecular programs for developmental progression.

To make the tissue-specific feature space directly inspectable, we ranked the genes in each tissue’s ordinal model by aggregate LightGBM importance and annotated the top 20 features per tissue via mygene.info and UniProt REST (Supplementary Table S7; row-z-scored trajectories shown in Supplementary Fig. S5). For the non-muscle models—brain, liver, lung, and blood—biologically coherent stage-defining genes emerge that were not previously discussed in the cross-species muscle module. In brain, the top annotated features include the ketogenic enzyme *HMGCS2* (peaking in infant) and the Hippo-pathway kinase *LATS1* (peaking post-pubertal), consistent with a metabolic-to-circuit-maturation transition during postnatal cortical development. In liver, *ALDH18A1* (proline/ornithine biosynthesis) and *COL1A1* peak in infancy alongside post-pubertal-peaking transcription factors *SATB2* and *CDH7*, mirroring the well-known amino-acid-anabolism-to-mature-hepatocyte program. In lung, the matricellular factor *CCN5* (post-pubertal; peak-vs-trough fold change 2.71) and the neural adhesion molecule *L1CAM* dominate alveolar maturation, whereas the oncofetal RNA-binder *IGF2BP2*—also a top cross-species muscle marker—signals neonatal mesenchymal growth, suggesting a shared early-postnatal proliferation programme across mesodermal tissues. In blood, ribosomal *RPLP1* (infant peak; fold change 3.14) and the chemokine receptor *CCR6* (post-pubertal) track the neonatal ribosomal/transcriptional surge and subsequent adaptive-immune maturation. We retained placeholder rows in Supplementary Table S7 for the top-ranked features that returned no curated description from either annotation source—predominantly novel pig loci—to flag

these as opportunities for follow-up functional annotation.

2.4 Evolutionary conservation with human development

A central question for the pig model is whether its developmental timing biologically mirrors that of humans. We focused cross-species validation on skeletal muscle because it is the most translationally relevant tissue in our atlas: porcine muscle models underpin research in Duchenne Muscular Dystrophy[28], sarcopenia, and meat quality, and musculoskeletal development is under strong biomechanical constraints that may promote evolutionary conservation. We performed a direct, quantitative comparison with the human skeletal muscle developmental RNA-seq dataset of Schaiter *et al.*[29] (Fig. 4).

We found significant evidence for conservation of developmental trajectories in muscle tissue. Using pre-specified significance and effect-size criteria (FDR < 0.10, Benjamini–Hochberg corrected, and $|\log_2 \text{FC}| > 0.5$ in both species; see Methods), we identified 66 genes with significant and biologically meaningful developmental fold-changes in both species. There was a significant positive correlation in fold-changes between species (Pearson $r = 0.69$, $p = 1.09 \times 10^{-10}$, 95% CI: 0.54–0.80; Supplementary Fig. S1), with 97.0% (64/66) of shared developmental markers changing in the same direction (Fig. 4a). Bootstrap analysis (1,000 iterations) confirmed the robustness of this correlation (95% CI: 0.50–0.82; Supplementary Fig. S1b). We note that the correlation magnitude decreases with more permissive FDR thresholds: sensitivity analysis across thresholds from 0.05 to 0.20 showed correlations ranging from $r = 0.61$ to $r = 0.69$ (Supplementary Fig. S1a), as more permissive thresholds include noisier genes.

We identified three conserved functional modules among the top cross-species markers driving this maturation (Fig. 4b,c; Supplementary Tables S3–S4).

Module 1: Neuromuscular Maturation and Growth Signalling. This module comprises genes with high infant expression that decrease during postnatal development, reflecting the transition from active myogenesis to mature muscle homeostasis. *ACHE* (pig FC = -3.34 , human FC = -0.53), essential for neuromuscular junction function[26], shows elevated expression in the infant stage (0–20 d; mean TPM = 17.5) that progressively declines to baseline in post-pubertal stages (TPM = 1.3). *SORCS2* (pig FC = -4.67 , human FC = -0.95), a regulator of motor neuron innervation[30], and *KREMEN1* (pig FC = -1.69 , human FC = -1.43), a Wnt/ β -catenin signalling antagonist[31, 32], show similar gradual postnatal declines. *NRK* (pig FC = -8.56 , human FC = -2.45), an embryonic Ste20-type kinase expressed predominantly in developing skeletal musculature[33], has the second-largest fold change and infant TPM = 128.2 declining to 0.3 in adults. The oncofetal RNA-binding protein *IGF2BP2* (pig FC = -2.19 , human FC = -2.03) regulates myoblast proliferation via translational control of growth-related transcripts[34]; the closely related *IGF2BP3* (pig FC = -4.85 , human FC = -1.23) shows a similar postnatal decline. Their downregulation accompanies the transition from proliferative to differentiating myoblasts. *ETV5* (pig FC = -2.80 , human FC = -2.24), a PEA3-subfamily transcription factor implicated in myogenic

differentiation and satellite cell activation[35]. *CCDC8* (pig FC = -4.44, human FC = -3.14), a core component of the 3M growth-regulatory complex, is implicated in a pathway controlling mammalian growth[36]; its downregulation parallels the transition from proliferative to hypertrophic tissue growth.

Module 2: Structural Refinement and ECM Remodelling. This module reflects the transition from rapid tissue growth to structural consolidation, encompassing both cytoskeletal dynamics and extracellular matrix maturation. *FGFRL1* (pig FC = -3.03, human FC = -2.10) is required for slow muscle fibre development[37]; its expression peaks during the infant stage (TPM = 77.4) and drops sharply in early childhood (TPM = 15.6). *DLK1* (pig FC = -9.15, human FC = -2.88) promotes myogenic differentiation during development[38, 39], with high infant expression (TPM = 338.8) declining sharply as the muscle fiber population matures. *POSTN* (pig FC = -7.97, human FC = -2.93), a secreted matricellular protein associated with extracellular matrix remodelling during myofibre differentiation and maturation[40], is the most highly expressed ECM gene in this set (infant TPM = 602.4), declining dramatically as tissue architecture stabilises. *ADAMTS2* (pig FC = -4.25, human FC = -1.83), the principal procollagen N-proteinase[41], and *COL11A2* (pig FC = -8.35, human FC = -0.70), a collagen component regulating fibril organisation[42], reflect the intensive ECM biosynthesis during rapid postnatal growth. *WDR1* (pig FC = -1.69, human FC = -0.52), a cofactor of cofilin-mediated actin filament disassembly with critical roles in muscle development[43], and *FSCN1* (pig FC = -4.28, human FC = -1.95), an actin-bundling protein driving cell migration during morphogenesis[44], capture the cytoskeletal remodelling required during myofibrillogenesis. *FREM2* (pig FC = -4.58, human FC = -1.05), a basement membrane protein of the FRAS/FREM complex[45], shows postnatal downregulation consistent with the transitional role of the FRAS/FREM complex during embryonic morphogenesis. *TCAF1* (pig FC = -4.72, human FC = -1.02), which regulates TRPM8 ion channel trafficking and calcium signalling[46], is the highest-importance gene among the cross-species markers (importance = 552.4), with infant TPM = 78.9 declining to 3.0 in adults. *TRIM54*, a muscle-specific E3 ubiquitin ligase that associates with the sarcomeric M-line and Z-line and is required for microtubule stability during myogenesis[47], shows the opposite pattern—lower infant expression (TPM = 78.3) increasing sharply in early childhood (TPM = 351.1) as the mature contractile apparatus is established.

Module 3: Metabolic Transition. This module captures genes upregulated during metabolic maturation of skeletal muscle. *ME1* (pig FC = 2.59, human FC = 0.64) encodes malic enzyme 1, a key source of NADPH for lipid biosynthesis[48]; its expression increases progressively from infant (TPM = 2.8) to adult stages (TPM = 16.9), reflecting the transition from carbohydrate to lipid metabolism. *MAOB* (pig FC = 2.55, human FC = 0.80), a mitochondrial outer membrane enzyme catalysing oxidative deamination of monoamines, shows concordant postnatal upregulation (infant TPM = 5.7 to adult TPM = 33.4), consistent with the general increase in mitochondrial content during postnatal muscle maturation. Notably, *PLA2G4E* (pig FC = 2.32, human FC = -1.25), a group IVE cytosolic phospholipase with

calcium-dependent N-acyltransferase activity that generates endocannabinoid precursors, is the only discordant gene among the top 20 markers, showing opposite developmental trajectories in pig and human—potentially reflecting species-specific differences in intramuscular lipid metabolism.

The 97% directional concordance among 66 independently identified genes—spanning neuromuscular junction maturation, extracellular matrix remodelling, and metabolic switching—is consistent with conservation of the core developmental logic of mammalian skeletal muscle over approximately 80 million years of divergence between *Sus scrofa* and *Homo sapiens*[49]. This supports the use of our transcriptomic atlas as a reference for porcine developmental staging. We note that the muscle cross-species comparison was limited to infant versus adult expression; a finer-grained alignment across all five porcine stages would require human developmental data with greater age resolution spanning childhood, adolescence, and young adulthood.

Cross-species expansion to additional tissues. To partially address the reviewer-recommended extension to non-muscle tissues, we re-applied the same conservation pipeline—strict one-to-one Ensembl Compara orthologs ($n = 7,485$ validated pairs), $\log_2$ fold-change between Infant and Adult in both species, Pearson correlation with bootstrap 95% CI—using the human developmental atlas of Cardoso-Moreira *et al.*[50] (E-MTAB-6814). One important data-availability caveat: the Cardoso-Moreira 2019 human RPKM matrix contains seven organs (Brain, Cerebellum, Heart, Kidney, Liver, Ovary, Testis) but *does not include lung*; consequently, the cross-species comparison for lung that the reviewer requested could not be performed with this dataset and is left as a future-work item that will require a dedicated human pulmonary developmental cohort. For the two reviewer-requested tissues that are present, the small per-tissue postnatal Cardoso-Moreira sample sizes (Brain: 7 young + 9 adult human samples; Liver: 4 young + 6 adult) yielded an underpowered human FDR test. We therefore additionally report a pig-anchored variant (genes passing the filter in pig only) and emphasise the strict variant where it has power. Brain showed a significant strict correlation (Pearson $r = 0.29$, 95% CI: 0.17–0.41, $p = 1.6 \times 10^{-6}$ over strict-filter orthologs, 85.3% directional concordance) and a smaller but highly significant pig-anchored correlation ($r = 0.21$, 95% CI: 0.16–0.25, $p = 2.7 \times 10^{-22}$, 2,193 genes). However, a closer inspection of the top brain strict-filter genes revealed that several of the largest infant-to-adult upregulated transcripts in the pig “brain” samples are canonical skeletal-muscle markers (e.g. *TNNC2*, *SMPX*, *MYBPHL*, *CHRNA1*, *LRRC2*); their median expression is near zero with high-TPM outliers in a small number of samples, suggesting variable muscle-tissue contamination in a subset of pig brain biopsies in PigGTEx. The brain cross-species correlation therefore partly reflects cross-species concordance of muscle developmental dynamics that “leak” into the brain expression matrix, and the strict-filter brain r should not be interpreted as a clean measure of brain-specific developmental conservation. Liver had no orthologs passing the strict bidirectional filter (consistent with the very small human cohort, $n = 10$); the pig-anchored estimate was

$r = 0.21$, 95% CI: 0.14–0.29, $p = 3.0 \times 10^{-10}$, 857 genes, 47.3% directional concordance—*below* the 50% chance baseline, meaning the pig-anchored liver analysis provides no evidence of cross-species concordance and is consistent with the strict null (Supplementary Fig. S7; Supplementary Table S8). Both extensions therefore underline that the reliable cross-species result is the one obtained for muscle with a matched human muscle cohort; obtaining clean per-tissue conservation estimates for brain, liver, and lung will require dedicated human postnatal cohorts with comparable size and matched tissue dissection rigour. We thank the reviewer for prompting this analysis, which we present in full—negatives included—in Supplementary Fig. S7 and Supplementary Table S8.

3 Discussion

We have established the first multi-tissue transcriptomic atlas for porcine development, providing a rigorous molecular alternative to subjective morphological staging[1]. By harmonising data across five tissues into a unified predictive framework, we show that biological maturity can be inferred from gene expression profiles with the held-out accuracies summarised in Fig. 2 (mean balanced accuracy 0.895; tissue-specific range 0.81–0.96). This shift from qualitative observation to quantitative measurement addresses a long-standing challenge in pig research: the standardisation of developmental staging across experimental cohorts and research groups. To facilitate community adoption, we provide an open-access interactive web application (<https://pigdevstage.streamlit.app>) that enables researchers to upload gene expression profiles and obtain developmental stage predictions with probability estimates.

Input requirements and preprocessing recommendations for the web application. The web application expects a CSV or TSV file containing a *samples* $\times$ *genes* (or *genes* $\times$ *samples*; orientation is auto-detected) expression matrix with *Sus scrofa* Ensembl gene identifiers in ENSSCG format; version suffixes (e.g. ENSSCG00000037539.2) are stripped automatically. The native input is raw TPM, which the app then $\log_2(x + 1)$ -transforms internally to match the training-time preprocessing; a checkbox allows users to indicate that the supplied values are already log-transformed. The classifier operates over the 31,908 protein-coding and lncRNA genes used during training; any genes missing from the user’s matrix are imputed as zero (unexpressed), which is appropriate for genes with low or no expression in the user’s library preparation but *not* for systematic feature absence (e.g. probe-set incompatibility). We therefore make three explicit usability recommendations: (i) for users whose RNA-seq data are annotated against gene symbols, RefSeq, or alternative Ensembl assemblies, the application also accepts gene symbols and resolves them to ENSSCG identifiers via `mygene.info`; we recommend that users verify the resulting gene-coverage fraction (reported in the app interface) is $\geq 80\%$ before trusting the prediction; (ii) for microarray data (e.g. Affymetrix Porcine Genome arrays), users should first convert probe-set IDs to Ensembl gene identifiers via the platform’s annotation file, aggregate multiple probes per gene

by mean or median, and treat the resulting intensities as expression values—in this regime, prediction confidence should be interpreted with caution because microarray dynamic range and normalisation differ from RNA-seq TPM; (iii) for cohorts dominated by an under-represented breed (notably Large white in muscle, where subgroup balanced accuracy was 0.61, $n = 68$; see Supplementary Fig. S8) we recommend re-calibration on a small in-house cohort before deploying the prediction in a critical experimental decision. These guidance items are also presented in the application’s in-context help text.

The practical implications for pig research are substantial. Our framework enables researchers to objectively classify animals by biological maturity rather than chronological age, reducing confounding variability in studies of nutrition, growth physiology, and disease susceptibility. For example, when evaluating dietary interventions or growth-promoting treatments, synchronising cohorts by molecular developmental stage rather than age should improve statistical power and reproducibility. The tissue-specific nature of our models also allows researchers to select staging approaches appropriate to their target organ system.

Our staging framework has immediate applications for disease modelling, particularly for age-dependent muscular disorders such as Duchenne Muscular Dystrophy (DMD)[28, 51]. Since the molecular programs for muscle development are partially conserved between pigs and humans, porcine models can provide superior physiological mirroring compared to rodents. DMD typically manifests in humans between 2–5 years of age[52], with rapid deterioration through the pre-teen years. Using our transcriptomic atlas, this developmental window corresponds approximately to infant stage (0–20 d) in pigs for early onset, with pathological changes becoming visible by pre-pubertal stage (60 d) and rapid changes occurring within post-pubertal stage (150 d). By providing a standardised molecular definition of developmental stages, the atlas enables researchers to identify specific developmentally conserved checkpoints that a successful gene therapy must restore. For example, researchers could track whether a gene therapy successfully stabilises *TRIM54* (microtubule stability) and *FGFRL1* (slow fibre development) expression back to healthy infant or early childhood levels. Critically, by using molecular markers to synchronise experimental cohorts rather than relying on chronological age alone, it may be possible to reduce variability in disease progression and treatment response, improving pre-clinical trial quality.

An important biological finding is the profound tissue-specificity of developmental markers. The extremely low cross-tissue feature overlap (0.2% Jaccard similarity) reflects genuine biological differences: each tissue uses distinct molecular programs for developmental progression, as evidenced by clear tissue clustering in expression space (Supplementary Fig. S2a). Within-tissue feature importance rankings showed moderate instability (Jaccard similarity 0.08–0.31; 6–42% of top 50 features stable across random seeds), which we interpret as biological gene redundancy: developmental genes operate in co-regulated modules, and tree-ensemble methods can select different members of the same functional module across runs while maintaining equivalent predictive performance (BA std < 0.06 across seeds for all tissues). This behaviour

parallels the well-characterised property of penalised regression in epigenetic clock construction, where multivariate optimisation routinely selects CpGs not individually correlated with age from co-regulated modules without compromising prediction accuracy[10]. This distinction between feature instability and prediction stability is important: the atlas produces consistent developmental staging regardless of which specific genes from redundant modules are selected.

To assess the biological relevance of our atlas, we performed a cross-species comparison with human skeletal muscle, identifying three shared regulatory modules—neuromuscular maturation and growth signalling, structural refinement and ECM remodelling, and metabolic transition—comprising 21 genes (including *NRK*, *POSTN*, *IGF2BP2/3*, and *SORCS2*) with conserved developmental trajectories. To our knowledge, the human dataset represents the most comprehensive publicly available postnatal skeletal muscle RNA-seq resource spanning infant-to-adult development ($n = 11$; Schaiter *et al.*[29]). Despite this constraint inherent to the field, the observed correlation ($r = 0.69$, 95% CI: 0.54–0.80) is statistically robust ($p = 1.09 \times 10^{-10}$), and the 97% directional concordance and the well-documented biological roles of the identified genes are consistent with our porcine atlas capturing developmental biology shared with humans rather than species-specific artefacts. Recent work on species-specific developmental tempo provides context for understanding why conserved developmental programs unfold at different rates across species[53]. However, we emphasise that cross-species validation is currently limited to muscle tissue; extending this analysis to brain, liver, lung, and blood represents a critical future direction that will require dedicated tissue-specific developmental RNA-seq datasets with sufficient postnatal sample sizes for gene-level statistical testing.

More broadly, our framework is designed to be species-agnostic: the only required inputs are bulk RNA-seq profiles with age metadata across developmental stages. Pan-mammalian epigenetic clocks have demonstrated that biological age prediction generalises across species, with a universal methylation clock now spanning 185 mammalian species[6–8]. While those approaches use methylation-based regression, our method applies ordinal machine learning classification to transcriptomic data; nevertheless, the underlying principle—extending biological age frameworks across species—is shared. The growing availability of multi-species developmental transcriptomic resources[15, 50] provides the data foundation for applying our transcriptomic staging approach to additional livestock and model species, enabling standardised developmental comparisons across agricultural and biomedical research.

We acknowledge the paradox of using chronologically defined bins to train a model intended to capture biological age. In the absence of a “gold standard” for physiological maturity, chronological age serves as a noisy proxy. Machine learning theory demonstrates that consistent learning is achievable even with noisy labels, provided appropriate methods are employed[54]; with sufficient data, the resulting model can approximate the true underlying signal—here, the molecular definition of a developmental stage—while being robust to individual labelling noise. Notably, ordinal classification is inherently more

robust to boundary noise than continuous age regression: misclassification between adjacent stages (e.g., pre-pubertal vs. post-pubertal) reflects genuine transitional biology rather than the regression-to-the-mean artefacts that confound absolute and relative age-acceleration estimates in continuous epigenetic clocks[10]. Consequently, discrepancies between model prediction and chronological age likely reflect true biological inter-individual variability (developmental acceleration or delay) rather than model error. The fact that the learned molecular signatures (e.g., *SORCS2*, *FGFRL1*) track development in humans, despite vastly different chronological lifespans, supports the interpretation that the model captures conserved biological programs rather than chronological associations alone.

To test whether the chronological-age boundaries we use correspond to genuine biological maturation switches, we performed an independent marker-gene validation. We curated a panel of 25 literature-supported postnatal mammalian markers spanning the four transition windows of our framework—suckling-to-weaning (~21 d; e.g. *AFP*, *ALB*, *IGF1*, *IGF2*, *LCT*), weaning-to-growth (~60 d; e.g. *MYH3*, *MYH8*, *MYH2*, *MSTN*, *IGFBP5*, *MBP*, *PLP1*, *SYP*), the pubertal switch (~150 d; e.g. *INSL3*, *LHB*, *AR*, *ESR1*, *GHR*), and adult maturation (~365 d; e.g. *CDKN2A*, *LMNA*, *SIRT1*, *SIRT3*, *MKI67*). For each marker $\times$ tissue pair we computed mean $\log_2(\text{TPM} + 1)$ per stage, fit one-way ANOVA with Tukey–HSD post-hoc, and identified the adjacent-stage pair carrying the largest mean difference. After excluding pairs lacking adequate per-stage coverage (≥ 5 samples per flanking stage), 30 marker $\times$ tissue combinations remained evaluable. Twenty of thirty (67%) matched the literature-predicted direction of change, and ten of thirty (33%) had their largest single-step change at the exact predicted transition—with T1 (suckling-to-weaning) showing the strongest window-level concordance (4/7 markers, 57%) (Supplementary Fig. S6). The most informative miss was the porcine myosin heavy-chain switch: *MYH3*, *MYH8* and *MYH2* showed correctly directed but earlier transitions (Infant $\rightarrow$ Early childhood) than the assumed weaning boundary, consistent with the rapid first-month perinatal isoform exchange documented in pig skeletal muscle. These observations are consistent with the five-stage chronological framework capturing genuine maturation switches at the perinatal, peri-weaning, and pubertal transitions, while clarifying that the calendar boundaries are approximations of underlying maturation programmes that overlap, and that some adjustments (such as an earlier myosin-switch midpoint) might better match the molecular dynamics in pig muscle.

Several limitations must be acknowledged. First, the present atlas is restricted to *postnatal* development (day 0 to >365 d). Mammalian development naturally spans prenatal (embryonic and fetal) and postnatal phases, and PigGTEx does not provide systematic multi-tissue prenatal RNA-seq sampling across these five tissues; we therefore make no claim about prenatal trajectories. Extending the framework to embryonic and fetal stages will require dedicated prospective sampling across pregnancy, and we anticipate that integration with single-cell atlases of porcine organogenesis will be the natural complement to our postnatal staging.

Second, the PigGTEx dataset aggregates RNA-seq profiles from multiple sequencing platforms and library protocols, which may introduce technical variation. No batch correction was applied prior to modelling because tree-based models are robust to monotonic platform shifts in expression. To directly assess whether cross-project technical variation confounds the developmental signals, we computed gene-level fold-changes within the two largest single-project muscle cohorts (PRJNA488311, $n_{\text{total}} = 81$; PRJNA486202, $n_{\text{total}} = 75$), each spanning four consecutive developmental stages under uniform protocols (see Methods). We repeated this analysis across three developmental contrasts (infant vs. early childhood, infant vs. pre-pubertal, and infant vs. post-pubertal). Within-project fold-changes correlated strongly with full-dataset estimates across all comparisons (pooled Pearson $r = 0.67\text{--}0.81$; Supplementary Table S5), and the 66 cross-species validated genes showed high agreement ($r = 0.76\text{--}0.98$), confirming that the developmental signatures reflect biology rather than batch artefacts even for the earliest and subtlest developmental transitions. As a more comprehensive assessment, we also performed an embedding-level batch-effect evaluation across all five tissues (Supplementary Fig. S4; Supplementary Table S6). For each tissue we projected the top 5,000 most variable genes into a 10-component PCA and a 2-D UMAP, and decomposed the variance of PC1–PC5 with respect to Stage and BioProject using ordinary least squares. Developmental stage alone explained a substantial fraction of PC1 variance in four of the five tissues: muscle $R^2_{\text{stage}} = 0.40$ (761 samples drawn from 56 BioProjects), brain $R^2_{\text{stage}} = 0.65$ (43 samples / 6 projects), lung $R^2_{\text{stage}} = 0.33$ (129 / 20), and blood $R^2_{\text{stage}} = 0.55$ (78 / 7). UMAPs of these tissues show visible infant-to-adult gradients even where several BioProjects co-occupy the same region of the embedding. Liver was the most batch-dominated tissue (PC1 $R^2_{\text{stage}} = 0.02$ vs $R^2_{\text{bioproject}} = 0.94$, 309 samples / 34 projects); however, a clear stage component is recovered at PC5 ($R^2_{\text{stage}} = 0.29$), and the held-out balanced accuracy for liver remains 0.921. We acknowledge openly that BioProject explains the larger absolute variance share in every tissue—this is expected because most PigGTEx projects sample a single narrow age window, so Stage and BioProject are partially confounded by design—and we therefore do not claim that the embedding is batch-free. Rather, the combined evidence (within-project muscle replication of full-dataset fold-changes, visible stage gradients in the UMAPs across tissues, robust held-out balanced accuracies of 0.81–0.96, and the cross-species correlation of $r = 0.69$ in muscle) supports the conclusion that the developmental signal captured by our models is biological, not driven by technical variation. Stratified K -fold cross-validation ensures all developmental stages are represented in both training and test sets, but does not test cross-project generalisation. Future work should prioritise prospective datasets where individual studies span multiple developmental stages, enabling evaluation of cross-project transferability.

Third, our cross-species validation uses a small human postnatal muscle developmental RNA-seq dataset ($n = 11$; 4 infants, 7 adults) spanning only two developmental windows—infant and adult—with a substantial age gap that omits critical developmental milestones including adolescence and puberty,

preventing a stage-by-stage alignment across all five porcine stages. The implications of this “infant-
vs-adult” gap deserve explicit discussion. The reported Pearson correlation ($r = 0.69$) summarises the
agreement of *net* infant-to-adult $\log_2$ fold changes; it does not test whether intermediate trajectories (the
early-childhood, pre-pubertal, or post-pubertal porcine stages) are themselves conserved. With childhood
and adolescent human data available, we would expect three possible outcomes for any given gene: (i) the
gene’s monotonic infant→adult direction is preserved through intermediate ages, which would strengthen
the conservation claim for that gene; (ii) the gene exhibits a transient peak or trough at an intermediate
human stage (e.g. around puberty), which would refine our pig↔human stage alignment rather than
overturn the directional concordance; or (iii) the trajectory shape diverges qualitatively, which would
identify species-specific re-routing of the developmental program. Because our current analysis cannot
distinguish these scenarios, we have framed all conservation claims in net infant-to-adult terms. Resolving
this will require human postnatal muscle RNA-seq cohorts with at least one childhood and one adolescent
sample group; until such datasets exist, our cross-species inferences should be read as evidence of *directional*
conservation of the overall maturation programme rather than of *trajectory-shape* conservation. To partially
address the tissue scope of this limitation, we also extended cross-species concordance analysis to brain,
liver, and lung using the developmental RNA-seq atlas of Cardoso-Moreira *et al.*[50] (Supplementary
Fig. S7; Supplementary Table S8), and we report the corresponding per-tissue correlations in Results 2.4.
Fourth, although the gene selection criteria ($\text{FDR} < 0.10$, $|\log_2 \text{FC}| > 0.5$) were pre-specified, sensitivity
analysis confirms robustness across a range of thresholds. Fifth, class imbalance necessitated reduced
classification schemes for some tissues (2-class binary maturity classification for brain, blood, and lung).
Sixth, within each chronological stage bin the cohort is not balanced for breed, sex, or contributing
BioProject. To quantify this and assess sensitivity to the resulting heterogeneity, we performed a stratified
subgroup audit (Supplementary Fig. S8; Supplementary Table S9): we replayed the same 5-fold stratified
outer split used in the main pipeline (seed=42) and refit each fold with the per-fold hyperparameters stored
alongside the published metrics, producing bit-identical cross-validated predictions, then recomputed
balanced accuracy within every Breed, Sex, and BioProject level satisfying $n \geq 10$. The audit confirms
the reviewer’s intuition. In our muscle data the dominant breed fraction per stage ranges from 28%
(Pre-pubertal) to 62% (Early childhood, Duroc), the Early childhood subset is 72% female (55/76), the
Adult stage is represented by only five samples of unknown breed, and per-breed muscle BA spans 0.61
(Large white, $n = 68$) to 0.97 (Duroc, $n = 53$). A within-stage label permutation test (1,000 shuffles)
rejects the chance null for muscle/Sex ($p = 0.005$), muscle/BioProject ($p = 0.015$), and liver/Breed
($p = 0.008$); muscle/Breed sits at the boundary of nominal significance ($p = 0.054$) and is therefore best
read as borderline rather than negative; Brain/Sex, Liver/Sex, and Lung/Sex are consistent with chance
($p \geq 0.23$). Critically, the overall point estimates remain within the reported range (Muscle 0.89, Liver
0.92, Brain 0.81, Lung 0.89, Blood 0.96) and no subgroup with $n \geq 50$ falls below 0.70 for muscle or liver.

The classifier should therefore be interpreted as calibrated to the breed/sex composition of PigGTE_x; deployment to cohorts dominated by under-represented breeds (e.g. exclusively Large white) warrants re-calibration, and we flag this explicitly in the web-application documentation. Seventh, the reliance on bulk RNA-seq averages signals across cell types. Future iterations incorporating single-cell resolution[55] will allow us to dissect precisely which cell populations drive these developmental clocks.

4 Conclusions

This work provides a transparent, reproducible molecular standard for developmental staging in pig research. The atlas offers an immediately applicable tool for standardising experimental cohorts in swine nutrition, production, and biomedical studies. To facilitate community adoption, we provide an open-access interactive web application (<https://pigdevstage.streamlit.app>) that enables researchers to upload gene expression profiles and obtain developmental stage predictions with probability estimates. As the pig continues to serve as a major species for both agricultural and biomedical research, this resource ensures that developmental age is defined with the precision required for rigorous, reproducible science.

5 Methods

5.1 Data acquisition and processing

Bulk RNA-seq transcript-per-million (TPM) matrices and metadata were obtained from the PigGTE_x resource[16, 17]. Ages recorded in days were harmonised to developmental stages: infant (0–20 d), early childhood (21–59 d), pre-pubertal (60–149 d), post-pubertal (150–365 d), and adult (>365 d). Stage boundaries were chosen to align with established physiological milestones in porcine development[1]: weaning onset (~21 d)[5], post-weaning transition completion (~60 d)[21], earliest puberty onset (~150 d; typical range 150–220 d)[22], and full maturity (>365 d). Preprocessing consisted of a $\log_2(x + 1)$ transformation applied to TPM values. Variance filtering and z -score standardisation were omitted because tree-based models are invariant to monotonic transformations and split-based feature selection is unaffected by feature scale.

5.2 Machine learning framework

We implemented a reduction-based ordinal classification framework using Light Gradient Boosting Machine (LightGBM)[56]. Given the ordinal nature of developmental stages ($y \in \{0, \dots, K - 1\}$), we decomposed the K -class problem into $K - 1$ independent binary classification subtasks (Frank & Hall reduction[57]), where the k -th subtask predicts the cumulative probability $\Pr(y > k \mid x)$ for $k \in \{0, \dots, K - 2\}$. The class probability for each ordinal label $c \in \{0, \dots, K - 1\}$ is recovered from successive differences of these

505 cumulative probabilities:

$$\Pr(y = c \mid x) = \Pr(y > c - 1 \mid x) - \Pr(y > c \mid x), \quad (1)$$

506 under the boundary conventions $\Pr(y > -1 \mid x) = 1$ and $\Pr(y > K - 1 \mid x) = 0$. Concretely,
 507 $\Pr(y = 0) = 1 - \Pr(y > 0)$, $\Pr(y = K - 1) = \Pr(y > K - 2)$, and intermediate classes are obtained directly
 508 from (1).

509 Models were trained on all available genes (31,908 per tissue), relying on LightGBM’s built-in
 510 regularisation for implicit feature selection rather than a separate pre-selection step. Nine hyperparameters
 511 were tuned automatically via Optuna[58]: `num_leaves`, `max_depth`, `min_data_in_leaf`, `learning_rate`,
 512 `n_estimators`, `feature_fraction`, `bagging_fraction`, λ_1 (L1), and λ_2 (L2). Class weighting was set to
 513 **balanced** and each binary sub-model used early stopping (`patience` = 10) to prevent overfitting.

514 Because each binary sub-model is trained independently, the raw cumulative-probability predictions
 515 are not guaranteed to be monotonically non-increasing in k , which would violate (1). We therefore
 516 did *not* use LightGBM’s `monotone_constraints` parameter (which constrains the sign of the partial
 517 derivative of an ensemble with respect to an input feature) and instead enforced ordinal consistency
 518 through two deterministic post-hoc projections of the predicted probabilities at inference time. First,
 519 the cumulative probabilities were clipped so that $\widehat{\Pr}(y > k) \leftarrow \min(\widehat{\Pr}(y > k), \widehat{\Pr}(y > k - 1))$ for $k \geq 1$,
 520 restoring the non-increasing ordering required by (1). Second, after differencing to obtain the class
 521 probabilities, we enforced unimodality by propagating non-increasing values away from the predicted
 522 mode in both directions, followed by clipping to $[0, 1]$ and row-wise renormalisation. Both projections
 523 are deterministic, are applied uniformly to training and held-out predictions, and are implemented in
 524 the `OrdinalLightGBM.predict_proba` method of the released codebase. This post-hoc enforcement of
 525 ordinal monotonicity is computationally cheap and preserves the calibration of the underlying binary
 526 classifiers.

527 Models were evaluated using nested cross-validation to ensure unbiased performance estimation. The
 528 outer loop employed stratified K -fold cross-validation ($K = 5$, automatically reduced for tissues with
 529 small class sizes). Within each outer fold, hyperparameters were optimised using an inner 3-fold stratified
 530 cross-validation loop with 25 Optuna trials per fold, guided by the Tree-structured Parzen Estimator
 531 (TPE) sampler and a median pruner. Held-out predictions from all outer folds were pooled for a single,
 532 unbiased estimate of generalisation performance. A final model was then retrained on all available data
 533 with hyperparameters tuned via a separate inner cross-validation study. Primary performance metrics
 534 included Balanced Accuracy (BA) and Macro-F1 score (Supplementary Fig. S3; Supplementary Tables S1
 535 and S2):

$$\text{BA} = \frac{1}{K} \sum_{c=0}^{K-1} \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \quad (2)$$

Confidence intervals (95%) were calculated using bootstrap resampling (1,000 iterations) on the pooled cross-validated predictions.

5.3 Cross-species evolutionary analysis

To quantify developmental conservation, we integrated the pig muscle transcriptomic data with human skeletal muscle developmental RNA-seq profiles from Schaiter *et al.*[29] (GEO accession GSE257558). Of the larger cohort described in that study, RNA-seq data were available for 11 subjects: 4 infants (ages 7–28 months) and 7 adults (ages 30–56 years), all from quadriceps femoris muscle biopsies. We computed the $\log_2$ fold change of gene expression between infant and adult stages for both species.

Ortholog filter (strict one-to-one). Pig (*Sus scrofa*) and human Ensembl gene identifiers were resolved to HGNC-style symbols via the `mygene.info` service; we retained only symbols appearing exactly once in each species’ annotation table and then verified each candidate pair against the Ensembl Compara REST homology endpoint, keeping only those whose reciprocal homology classification was `ortholog_one2one` (the Ensembl Compara confidence-1 equivalent for strict one-to-one orthologues). All cross-species analyses operate on this validated mapping (cached at `review/analyses/results/pig_human_one_to_one_orthologs.csv`).

Human dataset preprocessing. For the muscle comparison, raw RNA-seq read counts from Schaiter *et al.*[29] were converted to TPM, $\log_2(\text{TPM} + 1)$ -transformed, and restricted to the one-to-one ortholog set. For the multi-tissue cross-species extension (Supplementary Fig. S7; Supplementary Table S8) we used the Cardoso-Moreira *et al.* 2019 developmental atlas[50] (E-MTAB-6814): the published `Human.RPKM.tsv` matrix was filtered to postnatal samples by removing all columns whose stage label ends in `wpc`; remaining Cardoso-Moreira postnatal stage names were mapped to the porcine five-stage scheme (*newborn/infant/toddler* $\rightarrow$ Infant; *school/youngTeenager* $\rightarrow$ Pre-pubertal; *teenager/oldTeenager* $\rightarrow$ Post-pubertal; *youngAdult/youngMidAge* $\rightarrow$ Adult; *olderMidAge/senior* $\rightarrow$ Adult); $\log_2$ fold changes between Infant and Adult were computed on RPKM with a pseudocount of 0.01 (matching the pig TPM pipeline); per-gene significance was assessed with Welch’s *t*-test (more powerful than Mann–Whitney *U* at the small per-tissue postnatal *n* in this atlas), followed by Benjamini–Hochberg FDR correction; the gene universe was restricted to the validated one-to-one ortholog set above.

Conservation was defined as the Pearson correlation of these developmental fold-change trajectories. We applied pre-specified criteria to select genes: $\text{FDR} < 0.10$ (Benjamini–Hochberg corrected) and $|\log_2 \text{FC}| > 0.5$ in both species, retaining all genes passing these thresholds (66 genes in muscle). No additional selection by feature importance or top-*N* ranking was applied. We reported the Pearson correlation coefficient with 95% bootstrap confidence intervals (1,000 iterations). As a robustness check, we performed sensitivity analysis across FDR thresholds from 0.05 to 0.20, confirming consistent positive correlations across the parameter space (Supplementary Fig. S1a).

5.4 Within-project batch validation

To assess whether cross-project technical variation confounds developmental signals, we performed a within-project fold-change validation. We identified the two largest single-project muscle cohorts that each span multiple developmental stages under uniform protocols: PRJNA488311 ($n_{\text{total}} = 81$) and PRJNA486202 ($n_{\text{total}} = 75$), each containing samples from infant through post-pubertal stages. We computed fold-changes across three developmental contrasts, all using infant as the baseline: infant vs. early childhood, infant vs. pre-pubertal, and infant vs. post-pubertal. For each gene and each comparison, we computed $\log_2$ fold change (later stage / infant) and Mann–Whitney U p -value, with Benjamini–Hochberg FDR correction, following the same procedure as the cross-species analysis. We then correlated within-project fold-changes with full-dataset fold-changes using Pearson r , both genome-wide and for the 66 cross-species validated genes specifically. A pooled within-project estimate was computed by averaging the fold-changes from both projects before correlation. Bootstrap confidence intervals (1,000 iterations) were computed for the pooled correlation at each comparison (Supplementary Table S5).

5.5 Functional enrichment

Gene ontology (GO) enrichment of stage-informative markers was performed using g:Profiler[59], applying a Benjamini–Hochberg FDR correction ($p_{\text{adj}} < 0.05$). We focused on GO Biological Process (BP), KEGG, and Reactome pathway terms to identify physiological functions driving developmental transitions.

References

- [1] M. M. Swindle, A. Makin, A. J. Herron, F. J. Clubb, and K. S. Frazier. Swine as models in biomedical research and toxicology testing. *Vet Pathol*, 49(2):344–356, March 2012. ISSN 1544-2217. doi: 10.1177/0300985811402846.
- [2] Joan K. Lunney, Angelica Van Goor, Kristen E. Walker, Taylor Hailstock, Jasmine Franklin, and Chaohui Dai. Importance of the pig as a human biomedical model. *Science Translational Medicine*, 13(621):eabd5758, November 2021. doi: 10.1126/scitranslmed.abd5758. URL <https://www.science.org/doi/10.1126/scitranslmed.abd5758>.
- [3] Martien A. M. Groenen, Alan L. Archibald, Hirohide Uenishi, Christopher K. Tuggle, Yasuhiro Takeuchi, Max F. Rothschild, Claire Rogel-Gaillard, Chankyu Park, Denis Milan, Hendrik-Jan Megens, Shengting Li, Denis M. Larkin, Heebal Kim, Laurent A. F. Frantz, Mario Caccamo, Hyeonju Ahn, Bronwen L. Aken, Anna Anselmo, Christian Anthon, Loretta Auvil, Bouabid Badaoui, Craig W. Beattie, Christian Bendixen, Daniel Berman, Frank Blecha, Jonas Blomberg, Lars Bolund, Mirte Bosse, Sara Botti, Zhan Bujie, Megan Bystrom, Boris Capitanu, Denise Carvalho-Silva, Patrick Chardon, Celine Chen, Ryan Cheng, Sang-Haeng Choi, William Chow, Richard C. Clark, Christopher

Clee, Richard P. M. A. Crooijmans, Harry D. Dawson, Patrice Dehais, Fioravante De Sapio, Bert Dibbitts, Nizar Drou, Zhi-Qiang Du, Kellye Eversole, João Fadista, Susan Fairley, Thomas Faraut, Geoffrey J. Faulkner, Katie E. Fowler, Merete Fredholm, Eric Fritz, James G. R. Gilbert, Elisabetta Giuffra, Jan Gorodkin, Darren K. Griffin, Jennifer L. Harrow, Alexander Hayward, Kerstin Howe, Zhi-Liang Hu, Sean J. Humphray, Toby Hunt, Henrik Hornshøj, Jin-Tae Jeon, Patric Jern, Matthew Jones, Jerzy Jurka, Hiroyuki Kanamori, Ronan Kapetanovic, Jaebum Kim, Jae-Hwan Kim, Kyu-Won Kim, Tae-Hun Kim, Greger Larson, Kyooyeol Lee, Kyung-Tai Lee, Richard Leggett, Harris A. Lewin, Yingrui Li, Wansheng Liu, Jane E. Loveland, Yao Lu, Joan K. Lunney, Jian Ma, Ole Madsen, Katherine Mann, Lucy Matthews, Stuart McLaren, Takeya Morozumi, Michael P. Murtaugh, Jitendra Narayan, Dinh Truong Nguyen, Peixiang Ni, Song-Jung Oh, Suneel Onteru, Frank Panitz, Eung-Woo Park, Hong-Seog Park, Geraldine Pascal, Yogesh Paudel, Miguel Perez-Enciso, Ricardo Ramirez-Gonzalez, James M. Reecy, Sandra Rodriguez-Zas, Gary A. Rohrer, Lauretta Rund, Yongming Sang, Kyle Schachtschneider, Joshua G. Schraiber, John Schwartz, Linda Scobie, Carol Scott, Stephen Searle, Bertrand Servin, Bruce R. Southey, Goran Sperber, Peter Stadler, Jonathan V. Sweedler, Hakim Tafer, Bo Thomsen, Rashmi Wali, Jian Wang, Jun Wang, Simon White, Xun Xu, Martine Yerle, Guojie Zhang, Jianguo Zhang, Jie Zhang, Shuhong Zhao, Jane Rogers, Carol Churcher, and Lawrence B. Schook. Analyses of pig genomes provide insight into porcine demography and evolution. *Nature*, 491(7424):393–398, November 2012. ISSN 1476-4687. doi: 10.1038/nature11622.

- [4] Melanie E Boeyer, Emily V Leary, Richard J Sherwood, and Dana L Duren. Evidence of the non-linear nature of skeletal maturation. *Arch Dis Child*, 105(7):631–638, July 2020. ISSN 0003-9888. doi: 10.1136/archdischild-2019-317652. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC7592265/>.
- [5] John R. Pluske, Diana L. Turpin, and Jae-Cheol Kim. Gastrointestinal tract (gut) health in the young pig. *Anim Nutr*, 4(2):187–196, June 2018. ISSN 2405-6383. doi: 10.1016/j.aninu.2017.12.004.
- [6] A. T. Lu, Z. Fei, A. Haghani, T. R. Robeck, J. A. Zoller, C. Z. Li, R. Lowe, Q. Yan, J. Zhang, H. Vu, J. Ablaeva, V. A. Acosta-Rodriguez, D. M. Adams, J. Almunia, A. Aloysius, R. Ardehali, A. Arneson, C. S. Baker, G. Banks, K. Belov, N. C. Bennett, P. Black, D. T. Blumstein, E. K. Bors, C. E. Breeze, R. T. Brooke, J. L. Brown, G. G. Carter, A. Caulton, J. M. Cavin, L. Chakrabarti, I. Chatzistamou, H. Chen, K. Cheng, P. Chiavellini, O. W. Choi, S. M. Clarke, L. N. Cooper, M. L. Cossette, J. Day, J. DeYoung, S. DiRocco, C. Dold, E. E. Ehmke, C. K. Emmons, S. Emmrich, E. Erbay, C. Erlacher-Reid, C. G. Faulkes, S. H. Ferguson, C. J. Finno, J. E. Flower, J. M. Gaillard, E. Garde, L. Gerber, V. N. Gladyshev, V. Gorbunova, R. G. Goya, M. J. Grant, C. B. Green, E. N. Hales, M. B. Hanson, D. W. Hart, M. Haulena, K. Herrick, A. N. Hogan, C. J. Hogg, T. A. Hore, T. Huang, J. C. Izpisua Belmonte, A. J. Jasinska, G. Jones, E. Jourdain, O. Kashpur, H. Katcher, E. Katsumata, V. Kaza, H. Kiaris, M. S. Kobor, P. Kordowitzki, W. R. Koski, M. Krützen, S. B. Kwon, B. Larison, S. G. Lee, M. Lehmann, J. F. Lemaitre, A. J. Levine, C. Li, X. Li, A. R. Lim,

- D. T. S. Lin, D. M. Lindemann, T. J. Little, N. Macoretti, D. Maddox, C. O. Matkin, J. A. Mattison, M. McClure, J. Mergl, J. J. Meudt, G. A. Montano, K. Mozhui, J. Munshi-South, A. Naderi, M. Nagy, P. Narayan, P. W. Nathanielsz, N. B. Nguyen, C. Niehrs, J. K. O'Brien, P. O'Tierney Ginn, D. T. Odom, A. G. Ophir, S. Osborn, E. A. Ostrander, K. M. Parsons, K. C. Paul, M. Pellegrini, K. J. Peters, A. B. Pedersen, J. L. Petersen, D. W. Pietersen, G. M. Pinho, J. Plassais, J. R. Poganik, N. A. Prado, P. Reddy, B. Rey, B. R. Ritz, J. Robbins, M. Rodriguez, J. Russell, E. Rydkina, L. L. Sailer, A. B. Salmon, A. Sanghavi, K. M. Schachtschneider, D. Schmitt, T. Schmitt, L. Schomacher, L. B. Schook, K. E. Sears, A. W. Seifert, A. Seluanov, A. B. A. Shafer, D. Shanmuganayagam, A. V. Shindyapina, M. Simmons, K. Singh, I. Sinha, J. Slone, R. G. Snell, E. Soltanmaohammadi, M. L. Spangler, M. C. Spriggs, L. Staggs, N. Stedman, K. J. Steinman, D. T. Stewart, V. J. Sugrue, B. Szladovits, J. S. Takahashi, M. Takasugi, E. C. Teeling, M. J. Thompson, B. Van Bonn, S. C. Vernes, D. Villar, H. V. Vinters, M. C. Wallingford, N. Wang, R. K. Wayne, G. S. Wilkinson, C. K. Williams, R. W. Williams, X. W. Yang, M. Yao, B. G. Young, B. Zhang, Z. Zhang, P. Zhao, Y. Zhao, W. Zhou, J. Zimmermann, J. Ernst, K. Raj, and S. Horvath. Universal DNA methylation age across mammalian tissues. *Nat Aging*, 3(9):1144–1166, September 2023. ISSN 2662-8465. doi: 10.1038/s43587-023-00462-6. URL <https://www.nature.com/articles/s43587-023-00462-6>.
- [7] Kyle M. Schachtschneider, Lawrence B. Schook, Jennifer J. Meudt, Dhanansayan Shanmuganayagam, Joseph A. Zoller, Amin Haghani, Caesar Z. Li, Joshua Zhang, Andrew Yang, Ken Raj, and Steve Horvath. Epigenetic clock and DNA methylation analysis of porcine models of aging and obesity. *Geroscience*, 43(5):2467–2483, October 2021. ISSN 2509-2723. doi: 10.1007/s11357-021-00439-6.
- [8] Steve Horvath. DNA methylation age of human tissues and cell types. *Genome Biol*, 14(10):3156, December 2013. ISSN 1474-760X. doi: 10.1186/gb-2013-14-10-r115. URL <https://doi.org/10.1186/gb-2013-14-10-r115>.
- [9] Ran Duan, Qiaoyu Fu, Yu Sun, and Qingfeng Li. Epigenetic clock: A promising biomarker and practical tool in aging. *Ageing Res Rev*, 81:101743, November 2022. ISSN 1872-9649. doi: 10.1016/j.arr.2022.101743.
- [10] Andrew E. Teschendorff and Steve Horvath. Epigenetic ageing clocks: statistical methods and emerging computational challenges. *Nat Rev Genet*, 26(5):350–368, May 2025. ISSN 1471-0064. doi: 10.1038/s41576-024-00807-w. URL <https://www.nature.com/articles/s41576-024-00807-w>.
- [11] Akina Hoshino, Steve Horvath, Akshayalakshmi Sridhar, Alex Chitsazan, and Thomas A. Reh. Synchrony and asynchrony between an epigenetic clock and developmental timing. *Sci Rep*, 9(1):3770, March 2019. ISSN 2045-2322. doi: 10.1038/s41598-019-39919-3. URL <https://www.nature.com/articles/s41598-019-39919-3>.
- [12] Eric D. Sun, Olivia Y. Zhou, Max Hauptschein, Nimrod Rappoport, Lucy Xu, Paloma Navarro Negro, Ling Liu, Thomas A. Rando, James Zou, and Anne Brunet. Spatial transcriptomic clocks reveal

- cell proximity effects in brain ageing. *Nature*, 638(8049):160–171, February 2025. ISSN 1476-4687. doi: 10.1038/s41586-024-08334-8. URL <https://www.nature.com/articles/s41586-024-08334-8>.
- [13] Chandramouli Muralidharan, Enikő Zakar-Polyák, Anita Adami, Anna A. Abbas, Yogita Sharma, Raquel Garza, Jenny G. Johansson, Diahann A. M. Atacho, Éva Renner, Miklós Palkovits, Csaba Kerepesi, Johan Jakobsson, and Karolina Pircs. Human Brain Cell-Type-Specific Aging Clocks Based on Single-Nuclei Transcriptomics. *Advanced Science*, 12(43):e06109, 2025. ISSN 2198-3844. doi: 10.1002/advs.202506109. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/advs.202506109>. _eprint: <https://advanced.onlinelibrary.wiley.com/doi/pdf/10.1002/advs.202506109>.
- [14] BI Ageing Clock Team, Thomas M. Stubbs, Marc Jan Bonder, Anne-Katrien Stark, Felix Krueger, Ferdinand Von Meyenn, Oliver Stegle, and Wolf Reik. Multi-tissue DNA methylation age predictor in mouse. *Genome Biol*, 18(1):68, December 2017. ISSN 1474-760X. doi: 10.1186/s13059-017-1203-5. URL <http://genomebiology.biomedcentral.com/articles/10.1186/s13059-017-1203-5>.
- [15] Tim H. H. Coorens, Amy Guillaumet-Adkins, Rothem Kovner, Rebecca L. Linn, Victoria H. J. Roberts, Amrita Sule, and Patrick M. Van Hoose. The human and non-human primate developmental GTEx projects. *Nature*, 637(8046):557–564, January 2025. ISSN 1476-4687. doi: 10.1038/s41586-024-08244-9. URL <https://www.nature.com/articles/s41586-024-08244-9>.
- [16] Jinyan Teng, Yahui Gao, Hongwei Yin, Zhonghao Bai, Shuli Liu, Haonan Zeng, Lijing Bai, Zexi Cai, Bingru Zhao, Xiujin Li, Zhiting Xu, Qing Lin, Zhangyuan Pan, Wenjing Yang, Xiaoshan Yu, Dailu Guan, Yali Hou, Brittney N. Keel, Gary A. Rohrer, Amanda K. Lindholm-Perry, William T. Oliver, Maria Ballester, Daniel Crespo-Piazuelo, Raquel Quintanilla, Oriol Canela-Xandri, Konrad Rawlik, Charley Xia, Yuelin Yao, Qianyi Zhao, Wenye Yao, Liu Yang, Houcheng Li, Huicong Zhang, Wang Liao, Tianshuo Chen, Peter Karlsson-Mortensen, Merete Fredholm, Marcel Amills, Alex Clop, Elisabetta Giuffra, Jun Wu, Xiaodian Cai, Shuqi Diao, Xiangchun Pan, Chen Wei, Jinghui Li, Hao Cheng, Sheng Wang, Guosheng Su, Goutam Sahana, Mogens Sandø Lund, Jack C. M. Dekkers, Luke Kramer, Christopher K. Tuggle, Ryan Corbett, Martien A. M. Groenen, Ole Madsen, Marta Gòdia, Dominique Rocha, Mathieu Charles, Cong-jun Li, Hubert Pausch, Xiaoxiang Hu, Laurent Frantz, Yonglun Luo, Lin Lin, Zhongyin Zhou, Zhe Zhang, Zitao Chen, Leilei Cui, Ruidong Xiang, Xia Shen, Pinghua Li, Ruihua Huang, Guoqing Tang, Mingzhou Li, Yunxiang Zhao, Guoqiang Yi, Zhonglin Tang, Jicai Jiang, Fuping Zhao, Xiaolong Yuan, Xiaohong Liu, Yaosheng Chen, Xuewen Xu, Shuhong Zhao, Pengju Zhao, Chris Haley, Huaijun Zhou, Qishan Wang, Yuchun Pan, Xiangdong Ding, Li Ma, Jiaqi Li, Pau Navarro, Qin Zhang, Bingjie Li, Albert Tenesa, Kui Li, George E. Liu, Zhe Zhang, and Lingzhao Fang. A compendium of genetic regulatory effects across pig tissues. *Nat Genet*, 56(1):112–123, January 2024. ISSN 1546-1718. doi: 10.1038/s41588-023-01585-7. URL <https://www.nature.com/articles/s41588-023-01585-7>.
- [17] Lijuan Chen, Houcheng Li, Jinyan Teng, Zhen Wang, Xiaolu Qu, Zhe Chen, Xiaodian Cai, Haonan

- Zeng, Zhonghao Bai, Jinghui Li, Xiangchun Pan, Leyan Yan, Fei Wang, Lin Lin, Yonglun Luo, Goutam Sahana, Mogens Sandø Lund, Maria Ballester, Daniel Crespo-Piazuelo, Peter Karlskov-Mortensen, Merete Fredholm, Alex Clop, Marcel Amills, Crystal Loving, Christopher K. Tuggle, Ole Madsen, Jiaqi Li, Zhe Zhang, George E. Liu, Jicai Jiang, Lingzhao Fang, and Guoqiang Yi. Construction of a Multitissue Cell Atlas Reveals Cell-Type-Specific Regulation of Molecular and Complex Phenotypes in Pigs. *Adv Sci (Weinh)*, page e04961, November 2025. ISSN 2198-3844. doi: 10.1002/advs.202504961.
- [18] Jiajia Long, Weiwei Liu, Xinhao Fan, Yalan Yang, Xiaogan Yang, and Zhonglin Tang. A comprehensive atlas of pig RNA editome across 23 tissues reveals RNA editing affecting interaction mRNA-miRNAs. *G3 (Bethesda)*, 14(10):jkae178, October 2024. ISSN 2160-1836. doi: 10.1093/g3journal/jkae178.
- [19] The GTEx Consortium. The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science*, 369(6509):1318–1330, September 2020. doi: 10.1126/science.aaz1776. URL <https://www.science.org/doi/10.1126/science.aaz1776>.
- [20] Ryo Inoue, Takamitsu Tsukahara, Masako Nakatani, Mie Okutani, Ryoichiro Nishibayashi, Shohei Ogawa, Tomoko Harayama, Takayuki Nagino, Hironori Hatanaka, Kikuto Fukuta, Gustavo A. Romero-Pérez, Kazunari Ushida, and Denise Kelly. Weaning Markedly Affects Transcriptome Profiles and Peyer’s Patch Development in Piglet Ileum. *Front. Immunol.*, 6, December 2015. ISSN 1664-3224. doi: 10.3389/fimmu.2015.00630. URL <https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2015.00630/full>.
- [21] Colin Trengove Whittemore and I. Kyriazakis, editors. *Whittemore’s science and practice of pig production*. Blackwell Pub, Oxford, UK Ames, Iowa, 3rd ed edition, 2006. ISBN 978-1-4051-2448-5 978-0-470-99562-4 978-1-4051-7353-7. doi: 10.1002/9780470995624.
- [22] C. A. Lents, A. N. Lindo, S. M. Hileman, and D. J. Nonneman. Physiological and genomic insight into neuroendocrine regulation of puberty in gilts. *Domestic Animal Endocrinology*, 73: 106446, October 2020. ISSN 0739-7240. doi: 10.1016/j.domaniend.2020.106446. URL <https://www.sciencedirect.com/science/article/pii/S0739724020300138>.
- [23] Hyo Jung Kang, Yuka Imamura Kawasaki, Feng Cheng, Ying Zhu, Xuming Xu, Mingfeng Li, André M. M. Sousa, Mihovil Pletikos, Kyle A. Meyer, Goran Sedmak, Tobias Guennel, Yurae Shin, Matthew B. Johnson, Zeljka Krsnik, Simone Mayer, Sofia Fertuzinhos, Sheila Umlauf, Steven N. Lisgo, Alexander Vortmeyer, Daniel R. Weinberger, Shrikant Mane, Thomas M. Hyde, Anita Huttner, Mark Reimers, Joel E. Kleinman, and Nenad Sestan. Spatio-temporal transcriptome of the human brain. *Nature*, 478(7370):483–489, October 2011. ISSN 1476-4687. doi: 10.1038/nature10523.
- [24] Bridgette D. Semple, Klas Blomgren, Kayleen Gimlin, Donna M. Ferriero, and Linda J. Noble-Haeusslein. Brain development in rodents and humans: Identifying benchmarks of maturation and vulnerability to injury across species. *Prog Neurobiol*, 0:1–16, 2013. ISSN 0301-0082. doi:

- 10.1016/j.pneurobio.2013.04.001. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC3737272/>.
- [25] Hai Xia Zhang, Krzysztof Hyrc, and Liu Lin Thio. The glycine transport inhibitor sarcosine is an NMDA receptor co-agonist that differs from glycine. *J Physiol*, 587(Pt 13):3207–3220, July 2009. ISSN 1469-7793. doi: 10.1113/jphysiol.2009.168757.
- [26] Richard L. Rotundo. Expression and localization of acetylcholinesterase at the neuromuscular junction. *J Neurocytol*, 32(5):743–766, June 2003. ISSN 1573-7381. doi: 10.1023/B:NEUR.0000020621.58197.d4. URL <https://doi.org/10.1023/B:NEUR.0000020621.58197.d4>.
- [27] Anastasia Georgountzou and Nikolaos G. Papadopoulos. Postnatal Innate Immune Development: From Birth to Adulthood. *Front Immunol*, 8:957, August 2017. ISSN 1664-3224. doi: 10.3389/fimmu.2017.00957. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC5554489/>.
- [28] Nikolai Klymiuk, Andreas Blutke, Alexander Graf, Sabine Krause, Katinka Burkhardt, Annegret Wuensch, Stefan Krebs, Barbara Kessler, Valeri Zakhartchenko, Mayuko Kurome, Elisabeth Kemter, Hiroshi Nagashima, Benedikt Schoser, Nadja Herbach, Helmut Blum, Rüdiger Wanke, Annemieke Aartsma-Rus, Christian Thirion, Hanns Lochmüller, Maggie C. Walter, and Eckhard Wolf. Dystrophin-deficient pigs provide new insights into the hierarchy of physiological derangements of dystrophic muscle. *Hum Mol Genet*, 22(21):4368–4382, November 2013. ISSN 0964-6906. doi: 10.1093/hmg/ddt287. URL <https://doi.org/10.1093/hmg/ddt287>.
- [29] Alexander Schaiter, Andreas Hentschel, Felix Kleefeld, Julia Schuld, Vincent Umatham, Tara Procida-Kowalski, Christopher Nelke, Angela Roth, Andreas Hahn, Heidrun H. Krämer, Tobias Ruck, Rita Horvath, Peter F. M. van der Ven, Marek Bartkuhn, Andreas Roos, and Anne Schänzer. Molecular composition of skeletal muscle in infants and adults: a comparative proteomic and transcriptomic study. *Sci Rep*, 14(1):22965, October 2024. ISSN 2045-2322. doi: 10.1038/s41598-024-74913-4. URL <https://www.nature.com/articles/s41598-024-74913-4>.
- [30] Pernille Bogetofte Thomasen, Alena Salasova, Kasper Kjaer-Sorensen, Lucie Woloszczuková, Josef Lavický, Hande Login, Jeppe Tranberg-Jensen, Sergio Almeida, Sander Beel, Michaela Kavková, Per Qvist, Mads Kjolby, Peter Lund Ovesen, Stella Nolte, Benedicte Vestergaard, Andreea-Cornelia Udrea, Lene Niemann Nejsun, Moses V. Chao, Philip Van Damme, Jan Krivanek, Jeremy Dasen, Claus Oxvig, and Anders Nykjaer. SorCS2 binds progranulin to regulate motor neuron development. *Cell Reports*, 42(11):113333, November 2023. ISSN 2211-1247. doi: 10.1016/j.celrep.2023.113333. URL <https://www.sciencedirect.com/science/article/pii/S2211124723013451>.
- [31] Masako Osada, Emi Ito, Hector A. Fermin, Edwin Vazquez-Cintron, Tadmiri Venkatesh, Roland H. Friedel, and Mark Pezzano. The Wnt signaling antagonist Kremen1 is required for development of thymic architecture. *Clin Dev Immunol*, 13(2-4):299–319, 2006. ISSN 1740-2522. doi: 10.1080/17402520600935097.
- [32] Kevin Qin, Michael Yu, Jiaming Fan, Hongwei Wang, Piao Zhao, Guozhi Zhao, Wei Zeng, Connie

- Chen, Yonghui Wang, Annie Wang, Zander Schwartz, Jeffrey Hong, Lily Song, William Wagstaff, Rex C. Haydon, Hue H. Luu, Sherwin H. Ho, Jason Strelzow, Russell R. Reid, Tong-Chuan He, and Lewis L. Shi. Canonical and noncanonical Wnt signaling: Multilayered mediators, signaling mechanisms and major signaling crosstalk. *Genes & Diseases*, 11(1):103–134, January 2024. ISSN 2352-3042. doi: 10.1016/j.gendis.2023.01.030. URL <https://www.sciencedirect.com/science/article/pii/S2352304223000557>.
- [33] M. Kanai-Azuma, Y. Kanai, M. Okamoto, Y. Hayashi, H. Yonekawa, and K. Yazaki. Nr1: a murine X-linked NIK (Nck-interacting kinase)-related kinase gene expressed in skeletal muscle. *Mech Dev*, 89(1-2):155–159, December 1999. ISSN 0925-4773. doi: 10.1016/s0925-4773(99)00193-8.
- [34] Zhizhong Li, Jason A. Gilbert, Yunyu Zhang, Minsi Zhang, Qiong Qiu, Krishnan Ramanujan, Tea Shavlakadze, John K. Eash, Annarita Scaramozza, Matthew M. Goddeeris, David G. Kirsch, Kevin P. Campbell, Andrew S. Brack, and David J. Glass. An HMGA2-IGF2BP2 axis regulates myoblast proliferation and myogenesis. *Dev Cell*, 23(6):1176–1188, December 2012. ISSN 1878-1551. doi: 10.1016/j.devcel.2012.10.019.
- [35] J. M. Taylor, E. E. Dupont-Versteegden, J. D. Davies, J. A. Hassell, J. D. Houll , C. M. Gurley, and C. A. Peterson. A role for the ETS domain transcription factor PEA3 in myogenic differentiation. *Mol Cell Biol*, 17(9):5550–5558, September 1997. ISSN 0270-7306. doi: 10.1128/MCB.17.9.5550.
- [36] Dan Hanson, Philip G. Murray, James O’Sullivan, Jill Urquhart, Sarah Daly, Sanjeev S. Bhaskar, Leslie G. Biesecker, Mars Skae, Claire Smith, Trevor Cole, Jeremy Kirk, Kate Chandler, Helen Kingston, Dian Donnai, Peter E. Clayton, and Graeme C. M. Black. Exome Sequencing Identifies *CCDC8* Mutations in 3-M Syndrome, Suggesting that *CCDC8* Contributes in a Pathway with *CUL7* and *OBSL1* to Control Human Growth. *The American Journal of Human Genetics*, 89(1):148–153, July 2011. ISSN 0002-9297. doi: 10.1016/j.ajhg.2011.05.028. URL <https://www.sciencedirect.com/science/article/pii/S0002929711002217>.
- [37] Ruth Amann, Stefan Wyder, Anne M. Slavotinek, and Beat Trueb. The FgfrL1 receptor is required for development of slow muscle fibers. *Dev Biol*, 394(2):228–241, October 2014. ISSN 1095-564X. doi: 10.1016/j.ydbio.2014.08.016.
- [38] Jolena N. Waddell, Peijing Zhang, Yefei Wen, Sanjay K. Gupta, Aleksey Yevtodiyyenko, Jennifer V. Schmidt, Christopher A. Bidwell, Ashok Kumar, and Shihuan Kuang. Dlk1 is necessary for proper skeletal muscle development and regeneration. *PLoS One*, 5(11):e15055, November 2010. ISSN 1932-6203. doi: 10.1371/journal.pone.0015055.
- [39] Ditte Caroline Andersen, Jorge Laborda, Victoriano Baladron, Moustapha Kassem, S ren Paludan Sheikh, and Charlotte Harken Jensen. Dual role of delta-like 1 homolog (DLK1) in skeletal muscle development and adult muscle regeneration. *Development*, 140(18):3743–3753, September 2013. ISSN 1477-9129. doi: 10.1242/dev.095810.

- [40] Cansu Ozdemir, Uğur Akpulat, Parisa Sharafi, Yılmaz Yıldız, Ilyas Onbaşlar, and Cetin Kocaefe. Periostin is temporally expressed as an extracellular matrix component in skeletal muscle regeneration and differentiation. *Gene*, 553(2):130–139, December 2014. ISSN 1879-0038. doi: 10.1016/j.gene.2014.10.014.
- [41] A. Colige, S. W. Li, A. L. Sieron, B. V. Nusgens, D. J. Prockop, and C. M. Lapière. cDNA cloning and expression of bovine procollagen I N-proteinase: a new member of the superfamily of zinc-metalloproteinases with binding sites for cells and other matrix components. *Proc Natl Acad Sci U S A*, 94(6):2374–2379, March 1997. ISSN 0027-8424. doi: 10.1073/pnas.94.6.2374.
- [42] Mei Sun, Eric Y. Luo, Sheila M. Adams, Thomas Adams, Yaping Ye, Snehal S. Shetye, Louis J. Soslowsky, and David E. Birk. Collagen XI regulates the acquisition of collagen fibril structure, organization and functional properties in tendon. *Matrix Biol*, 94:77–94, December 2020. ISSN 0945-053X. doi: 10.1016/j.matbio.2020.09.001. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC7722227/>.
- [43] Shoichiro Ono. Functions of actin-interacting protein 1 (AIP1)/WD repeat protein 1 (WDR1) in actin filament dynamics and cytoskeletal regulation. *Biochem Biophys Res Commun*, 506(2):315–322, November 2018. ISSN 1090-2104. doi: 10.1016/j.bbrc.2017.10.096.
- [44] Josephine C. Adams. Roles of fascin in cell adhesion and motility. *Curr Opin Cell Biol*, 16(5):590–596, October 2004. ISSN 0955-0674. doi: 10.1016/j.ceb.2004.07.009.
- [45] Daiji Kiyozumi, Nagisa Sugimoto, and Kiyotoshi Sekiguchi. Breakdown of the reciprocal stabilization of QBRICK/Frem1, Fras1, and Frem2 at the basement membrane provokes Fraser syndrome-like defects. *Proc Natl Acad Sci U S A*, 103(32):11981–11986, August 2006. ISSN 0027-8424. doi: 10.1073/pnas.0601011103.
- [46] Dimitra Gkika, Loic Lemonnier, George Shapovalov, Dmitri Gordienko, Céline Poux, Michela Bernardini, Alexandre Bokhobza, Gabriel Bidaux, Cindy Degerny, Kathye Verreman, Basma Guarmit, Mohamed Benahmed, Yvan de Launoit, Rene J. M. Bindels, Alessandra Fiorio Pla, and Natalia Prevarskaya. TRP channel-associated factors are a novel protein family that regulates TRPM8 trafficking and activity. *J Cell Biol*, 208(1):89–107, January 2015. ISSN 1540-8140. doi: 10.1083/jcb.201402076.
- [47] Carol C. Gregorio, Cynthia N. Perry, and Abigail S. McElhinny. Functional properties of the titin/connectin-associated proteins, the muscle-specific RING finger proteins (MURFs), in striated muscle. *J Muscle Res Cell Motil*, 26(6):389–400, December 2005. ISSN 1573-2657. doi: 10.1007/s10974-005-9021-x. URL <https://doi.org/10.1007/s10974-005-9021-x>.
- [48] Frank A. Simmen, Iad Alhallak, and Rosalia C. M. Simmen. Malic enzyme 1 (ME1) in the biology of cancer: it is not just intermediary metabolism. *Journal of Molecular Endocrinology*, 65(4):

- R77–R90, November 2020. ISSN 1479-6813, 0952-5041. doi: 10.1530/JME-20-0176. URL <https://jme.bioscientifica.com/view/journals/jme/65/4/JME-20-0176.xml>.
- [49] Sudhir Kumar, Michael Suleski, Jack M. Craig, Adrienne E. Kasprowicz, Maxwell Sanderford, Michael Li, Glen Stecher, and S. Blair Hedges. TimeTree 5: An Expanded Resource for Species Divergence Times. *Mol Biol Evol*, 39(8):msac174, August 2022. ISSN 1537-1719. doi: 10.1093/molbev/msac174.
- [50] Margarida Cardoso-Moreira, Jean Halbert, Delphine Valloton, Britta Velten, Chunyan Chen, Yi Shao, Angélica Liechti, Kelly Ascensão, Coralie Rummel, Svetlana Ovchinnikova, Pavel V. Mazin, Ioannis Xenarios, Keith Harshman, Matthew Mort, David N. Cooper, Carmen Sandi, Michael J. Soares, Paula G. Ferreira, Sandra Afonso, Miguel Carneiro, James M. A. Turner, John L. VandeBerg, Amir Fallahshahroudi, Per Jensen, Rüdiger Behr, Steven Lisgo, Susan Lindsay, Philipp Khaitovich, Wolfgang Huber, Julie Baker, Simon Anders, Yong E. Zhang, and Henrik Kaessmann. Gene expression across mammalian organ development. *Nature*, 571(7766):505–509, July 2019. ISSN 1476-4687. doi: 10.1038/s41586-019-1338-5.
- [51] J. T. Selsby, J. W. Ross, D. Nonneman, and K. Hollinger. Porcine Models of Muscular Dystrophy. *ILAR Journal*, 56(1):116–126, May 2015. ISSN 1084-2020. doi: 10.1093/ilar/ilv015. URL <https://academic.oup.com/ilarjournal/article-lookup/doi/10.1093/ilar/ilv015>.
- [52] Katharine Bushby, Richard Finkel, David J. Birnkrant, Laura E. Case, Paula R. Clemens, Linda Cripe, Ajay Kaul, Kathi Kinnett, Craig McDonald, Shree Pandya, James Poysky, Frederic Shapiro, Jean Tomezsko, Carolyn Constantin, and DMD Care Considerations Working Group. Diagnosis and management of Duchenne muscular dystrophy, part 1: diagnosis, and pharmacological and psychosocial management. *Lancet Neurol*, 9(1):77–93, January 2010. ISSN 1474-4465. doi: 10.1016/S1474-4422(09)70271-6.
- [53] Ryohei Iwata and Pierre Vanderhaeghen. Metabolic mechanisms of species-specific developmental tempo. *Developmental Cell*, 59(13):1628–1639, July 2024. ISSN 1534-5807. doi: 10.1016/j.devcel.2024.05.027. URL <https://www.sciencedirect.com/science/article/pii/S1534580724003617>.
- [54] Nagarajan Natarajan, Inderjit S Dhillon, Pradeep K Ravikumar, and Ambuj Tewari. Learning with Noisy Labels. In *Advances in Neural Information Processing Systems*, volume 26. Curran Associates, Inc., 2013. URL https://proceedings.neurips.cc/paper_files/paper/2013/hash/3871bd64012152bfb53fdf04b401193f-Abstract.html.
- [55] Nicholas Schaum, Jim Karkanas, Norma F. Neff, Andrew P. May, Stephen R. Quake, Tony Wyss-Coray, Spyros Darmanis, Joshua Batson, Olga Botvinnik, Michelle B. Chen, Steven Chen, Foad Green, Robert Jones, Ashley Maynard, Lolita Penland, Angela Oliveira Pisco, Rene V. Sit, Geoffrey M. Stanley, James T. Webber, Fabio Zanini, Ankit S Baghel, Isaac Bakerman, Ishita Bansal, Daniela Berdnik, Biter Bilen, Douglas Brownfield, Corey Cain, Michelle B. Chen, Steven Chen, Min Cho, Giana Cirolia, Stephanie D. Conley, Spyros Darmanis, Aaron Demers, Kubilay Demir, Antoine de Morree,

Tessa Divita, Haley du Bois, Laughing Bear Torrez Dulgeroff, Hamid Ebadi, F. Hernán Espinoza,
 Matt Fish, Qiang Gan, Benson M. George, Astrid Gillich, Foad Green, Geraldine Genetiano, Xueying
 Gu, Gunsagar S. Gulati, Yan Hang, Shayan Hosseinzadeh, Albin Huang, Tal Iram, Taichi Isobe,
 Feather Ives, Robert Jones, Kevin S. Kao, Guruswamy Karnam, Aaron M. Kershner, Bernhard M.
 Kiss, William Kong, Maya E. Kumar, Jonathan Lam, Davis P. Lee, Song E. Lee, Guang Li, Qingyun
 Li, Ling Liu, Annie Lo, Wan-Jin Lu, Anoop Manjunath, Andrew P. May, Kaia L. May, Oliver L.
 May, Ashley Maynard, Marina McKay, Ross J. Metzger, Marco Mignardi, Dullei Min, Ahmad N.
 Nabhan, Norma F. Neff, Katharine M. Ng, Joseph Noh, Rasika Patkar, Weng Chuan Peng, Lolita
 Penland, Robert Puccinelli, Eric J. Rulifson, Nicholas Schaum, Shaheen S. Sikandar, Rahul Sinha,
 Rene V. Sit, Krzysztof Szade, Weilun Tan, Cristina Tato, Krissie Tellez, Kyle J. Travaglini, Carolina
 Tropini, Lucas Waldburger, Linda J. van Weele, Michael N. Wosczyzna, Jinyi Xiang, Soso Xue,
 Justin Youngyunpipatkul, Fabio Zanini, Macy E. Zardeneta, Fan Zhang, Lu Zhou, Ishita Bansal,
 Steven Chen, Min Cho, Giana Cirolia, Spyros Darmanis, Aaron Demers, Tessa Divita, Hamid
 Ebadi, Geraldine Genetiano, Foad Green, Shayan Hosseinzadeh, Feather Ives, Annie Lo, Andrew P.
 May, Ashley Maynard, Marina McKay, Norma F. Neff, Lolita Penland, Rene V. Sit, Weilun Tan,
 Lucas Waldburger, Justin Youngyunpipatkul, Joshua Batson, Olga Botvinnik, Paola Castro, Derek
 Croote, Spyros Darmanis, Joseph L. DeRisi, Jim Karkanias, Angela Oliveira Pisco, Geoffrey M.
 Stanley, James T. Webber, Fabio Zanini, Ankit S. Baghel, Isaac Bakerman, Joshua Batson, Biter
 Bilen, Olga Botvinnik, Douglas Brownfield, Michelle B. Chen, Spyros Darmanis, Kubilay Demir,
 Antoine de Morree, Hamid Ebadi, F. Hernán Espinoza, Matt Fish, Qiang Gan, Benson M. George,
 Astrid Gillich, Xueying Gu, Gunsagar S. Gulati, Yan Hang, Albin Huang, Tal Iram, Taichi Isobe,
 Guruswamy Karnam, Aaron M. Kershner, Bernhard M. Kiss, William Kong, Christin S. Kuo,
 Jonathan Lam, Benoit Lehallier, Guang Li, Qingyun Li, Ling Liu, Wan-Jin Lu, Dullei Min, Ahmad N.
 Nabhan, Katharine M. Ng, Patricia K. Nguyen, Rasika Patkar, Weng Chuan Peng, Lolita Penland,
 Eric J. Rulifson, Nicholas Schaum, Shaheen S. Sikandar, Rahul Sinha, Krzysztof Szade, Serena Y.
 Tan, Krissie Tellez, Kyle J. Travaglini, Carolina Tropini, Linda J. van Weele, Bruce M. Wang,
 Michael N. Wosczyzna, Jinyi Xiang, Hanadie Yousef, Lu Zhou, Joshua Batson, Olga Botvinnik,
 Steven Chen, Spyros Darmanis, Foad Green, Andrew P. May, Ashley Maynard, Angela Oliveira
 Pisco, Stephen R. Quake, Nicholas Schaum, Geoffrey M. Stanley, James T. Webber, Tony Wyss-
 Coray, Fabio Zanini, Philip A. Beachy, Charles K. F. Chan, Antoine de Morree, Benson M. George,
 Gunsagar S. Gulati, Yan Hang, Kerwyn Casey Huang, Tal Iram, Taichi Isobe, Aaron M. Kershner,
 Bernhard M. Kiss, William Kong, Guang Li, Qingyun Li, Ling Liu, Wan-Jin Lu, Ahmad N. Nabhan,
 Katharine M. Ng, Patricia K. Nguyen, Weng Chuan Peng, Eric J. Rulifson, Nicholas Schaum,
 Shaheen S. Sikandar, Rahul Sinha, Krzysztof Szade, Kyle J. Travaglini, Carolina Tropini, Bruce M.
 Wang, Kenneth Weinberg, Michael N. Wosczyzna, Sean M. Wu, Hanadie Yousef, Ben A. Barres,

- Philip A. Beachy, Charles K. F. Chan, Michael F. Clarke, Spyros Darmanis, Kerwyn Casey Huang, Jim Karkanias, Seung K. Kim, Mark A. Krasnow, Maya E. Kumar, Christin S. Kuo, Andrew P. May, Ross J. Metzger, Norma F. Neff, Roel Nusse, Patricia K. Nguyen, Thomas A. Rando, Justin Sonnenburg, Bruce M. Wang, Kenneth Weinberg, Irving L. Weissman, Sean M. Wu, Stephen R. Quake, and Tony Wyss-Coray. Single-cell transcriptomics of 20 mouse organs creates a Tabula Muris. *Nature*, 562(7727):367–372, October 2018. ISSN 0028-0836. doi: 10.1038/s41586-018-0590-4. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC6642641/>.
- [56] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html.
- [57] Eibe Frank and Mark Hall. A Simple Approach to Ordinal Classification. In *Proceedings of the 12th European Conference on Machine Learning*, EMCL ’01, pages 145–156, Berlin, Heidelberg, 2001. Springer-Verlag. ISBN 978-3-540-42536-6.
- [58] Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A Next-generation Hyperparameter Optimization Framework, July 2019. URL <http://arxiv.org/abs/1907.10902>. arXiv:1907.10902 [cs].
- [59] Liis Kolberg, Uku Raudvere, Ivan Kuzmin, Priit Adler, Jaak Vilo, and Hedi Peterson. g:Profiler-interoperable web service for functional enrichment analysis and gene identifier mapping (2023 update). *Nucleic Acids Res*, 51(W1):W207–W212, July 2023. ISSN 1362-4962. doi: 10.1093/nar/gkad347.

Declarations

Ethics approval and consent to participate

Not applicable. This study used publicly available transcriptomic data from the PigGTEx resource and did not involve any new animal experiments or human participants.

Consent for publication

Not applicable.

Availability of data and materials

The PigGTEx data used in this study is available at <https://piggtx.farmgtex.org/>. Raw and processed human RNA-seq data are available in the NCBI Gene Expression Omnibus (GEO) under

945 accession number GSE257558. All code required to reproduce the findings is available at [https://github.](https://github.com/TianYuan-Liu/pig-dev-stage)
946 [com/TianYuan-Liu/pig-dev-stage](https://github.com/TianYuan-Liu/pig-dev-stage). An interactive web application for developmental stage prediction
947 is available at <https://pigdevstage.streamlit.app>. All analyses were conducted using Python 3.9 and
948 R 4.2.0.

949 **Competing interests**

950 The authors declare no competing interests.

951 **Funding**

952 T.L. was funded by the European Union Marie Skłodowska-Curie Actions Doctoral Network project
953 LongTREC (HORIZON-MSCA-2021-DN-01 grant agreement project 101072892).

954 **Authors' contributions**

955 T.L. conceived the study, designed the methodology, performed all computational analyses, interpreted
956 results, and wrote the manuscript. R.L. and I.M.K. contributed to manuscript editing and review. P.T.
957 supervised the study, provided guidance, and edited the manuscript. All authors reviewed and approved
958 the final manuscript.

959 **Acknowledgements**

960 We thank the PigGTEx consortium for providing the transcriptomic data that made this study possible.

961 **Authors' information**

962 Not applicable.

Figures

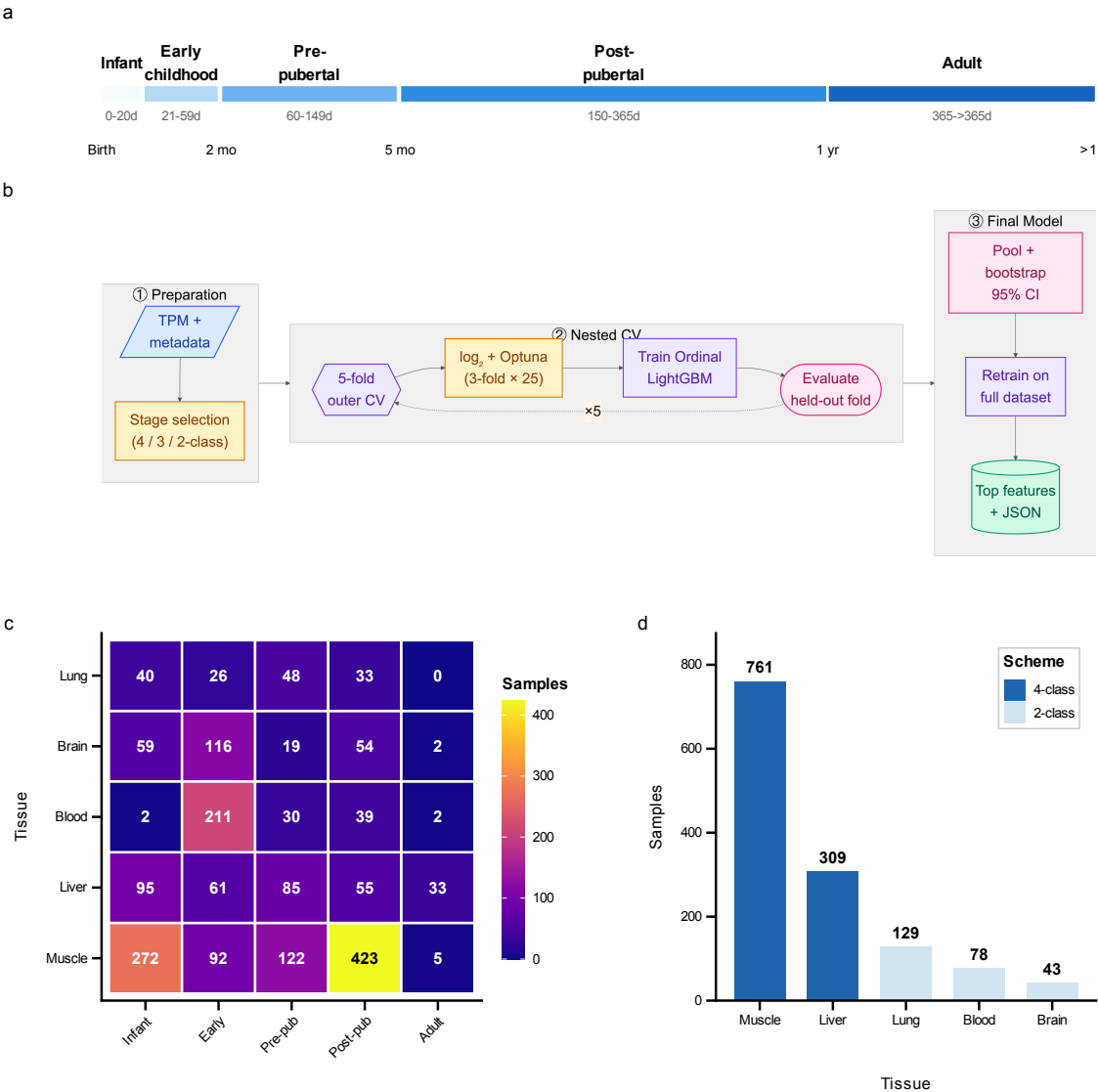


Figure 1: Study design and data landscape. **a**, Developmental stage definitions mapping chronological porcine age to five ordered stages: infant (0–20 d), early childhood (21–59 d), pre-pubertal (60–149 d), post-pubertal (150–365 d), and adult (>365 d). **b**, Machine learning workflow diagram illustrating the pipeline from data preprocessing to model calibration and evaluation. **c**, Sample distribution heatmap showing tissue-by-stage availability across 1,924 samples from five tissues. **d**, Classification scheme assignments based on per-class sample availability, with 4-class schemes for muscle and liver, and 2-class schemes for blood, brain, and lung.

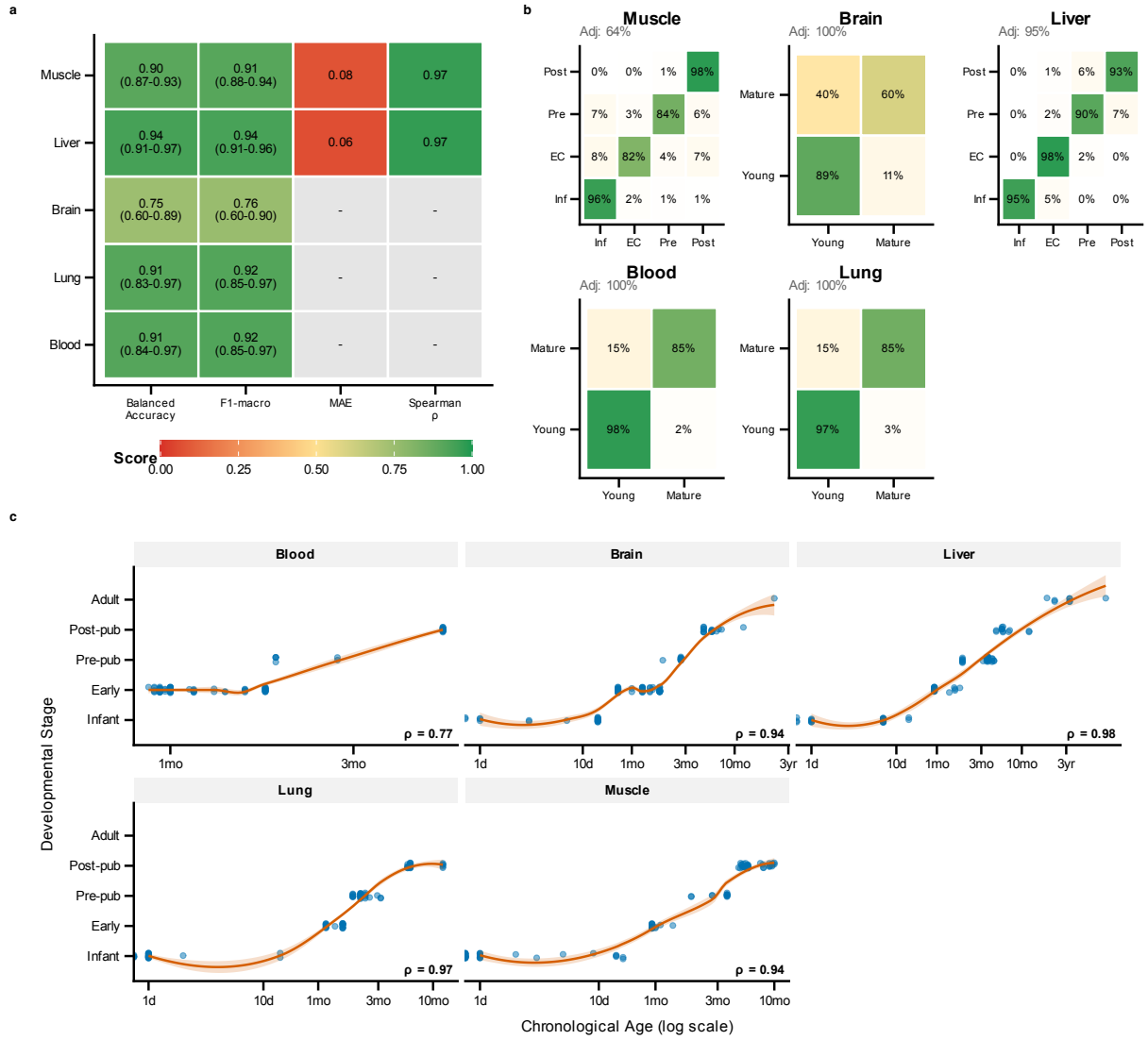


Figure 2: Classifier performance diagnostics. **a**, Performance metrics heatmap showing balanced accuracy, macro F1, mean absolute error, and Spearman correlations with 95% bootstrap confidence intervals across porcine tissues. Mean balanced accuracy across all tissues is 0.895, ranging from 0.811 (brain) to 0.959 (blood). **b**, Confusion matrices for each tissue showing classification accuracy and error patterns. **c**, Scatter plots of predicted developmental stage versus chronological age, demonstrating strong age-stage correlations (muscle $\rho = 0.965$, liver $\rho = 0.949$).

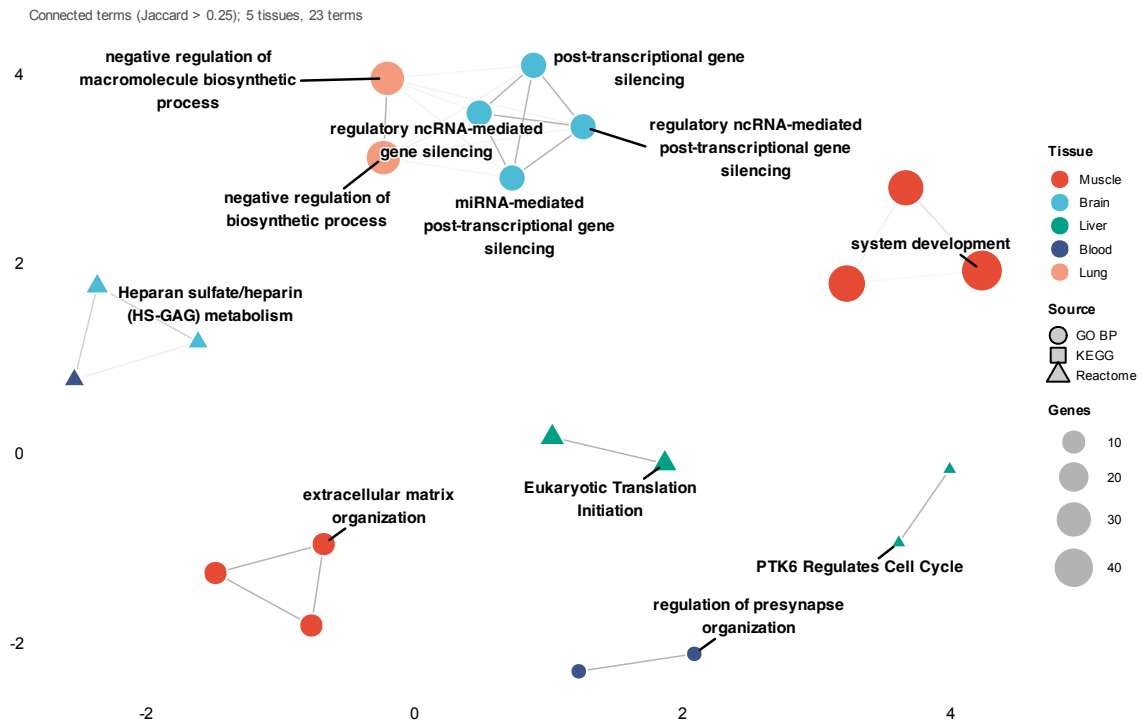


Figure 3: Functional enrichment analysis of stage-informative porcine genes. Functional network showing interconnected enrichment modules, with nodes representing enriched terms coloured by tissue and edges connecting terms sharing >25% of genes (Jaccard similarity). Hubs and cluster representatives are labelled with biological process or pathway names.

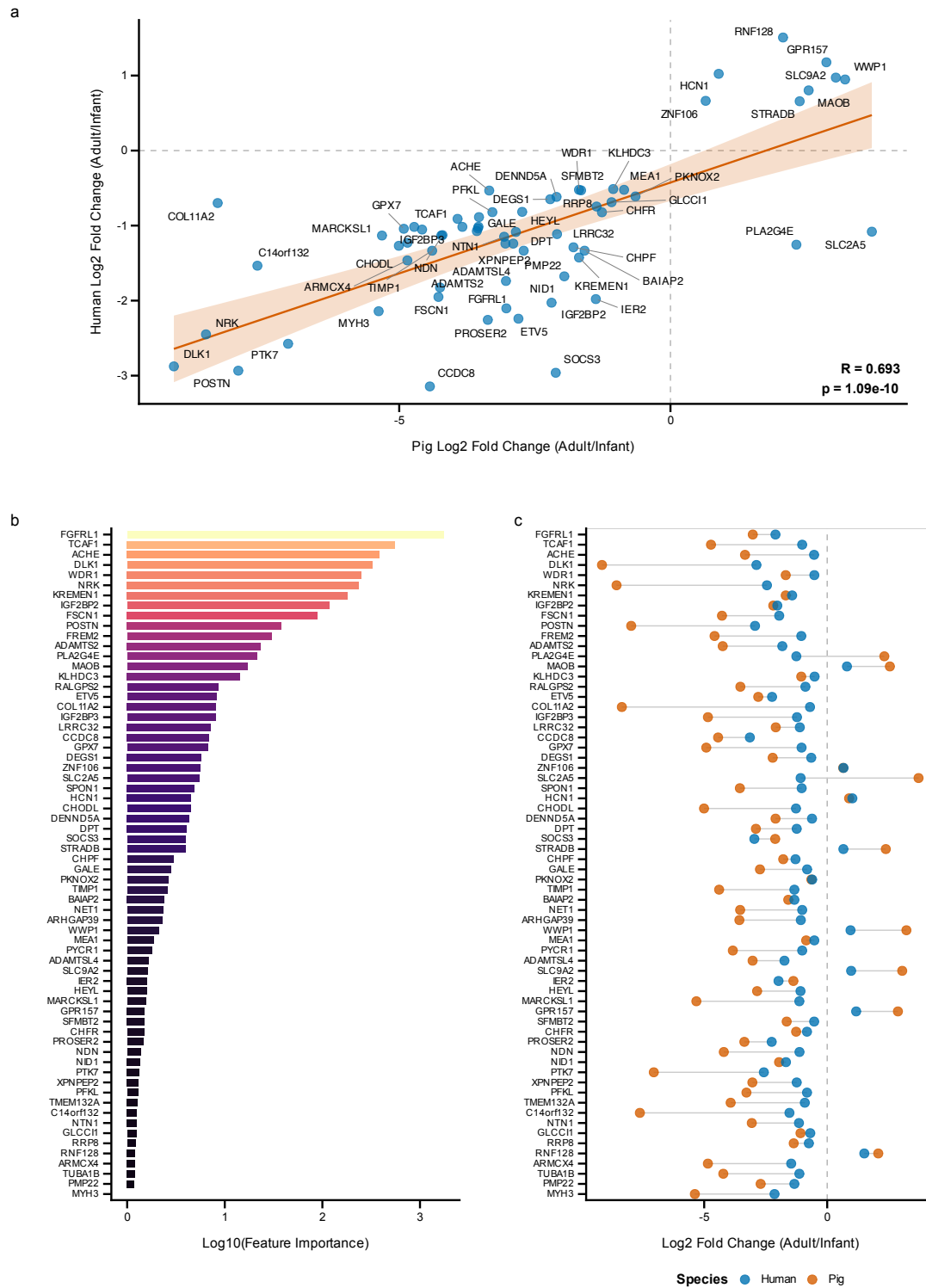


Figure 4: Cross-species validation of developmental gene expression. Human skeletal muscle transcriptomic data from Schaiter *et al.*[29] comparing infant (7–28 months) versus adult (30–56 years) samples were used to validate conservation of developmental signatures. For porcine samples, the comparison uses infant stage (0–20 d) versus post-pubertal/adult stage (>150 d). The five porcine developmental stages are: infant (0–20 d), early childhood (21–59 d), pre-pubertal (60–149 d), post-pubertal (150–365 d), and adult (>365 d). **a**, Scatter plot of log₂ fold changes between pig and human, showing significant positive correlation (Pearson $r = 0.693$, $p = 1.09 \times 10^{-10}$, 95% CI: 0.542–0.801). Of 66 genes meeting pre-specified criteria (FDR < 0.10, Benjamini–Hochberg corrected, and $|\log_2 FC| > 0.5$) in both species, 64 (97.0%) showed conserved direction of change. **b**, Feature importance of top conserved markers in the developmental stage classifier. **c**, Paired fold change comparison showing pig and human values for each conserved gene. Top markers include *FGFR1*, *DLK1*, *KREMEN1*, *ACHE*, and *TCAF1*.